

The background of the image is a deep black space filled with a vast number of stars. The stars vary in size, brightness, and color, with many appearing as small white or yellow dots, while others are larger and more distinct, showing blue or orange hues. Some stars have prominent diffraction spikes, giving them a cross-like appearance. A semi-transparent dark grey rectangular box is centered horizontally and vertically, containing the text "Star Formation" in a bright yellow, sans-serif font.

Star Formation

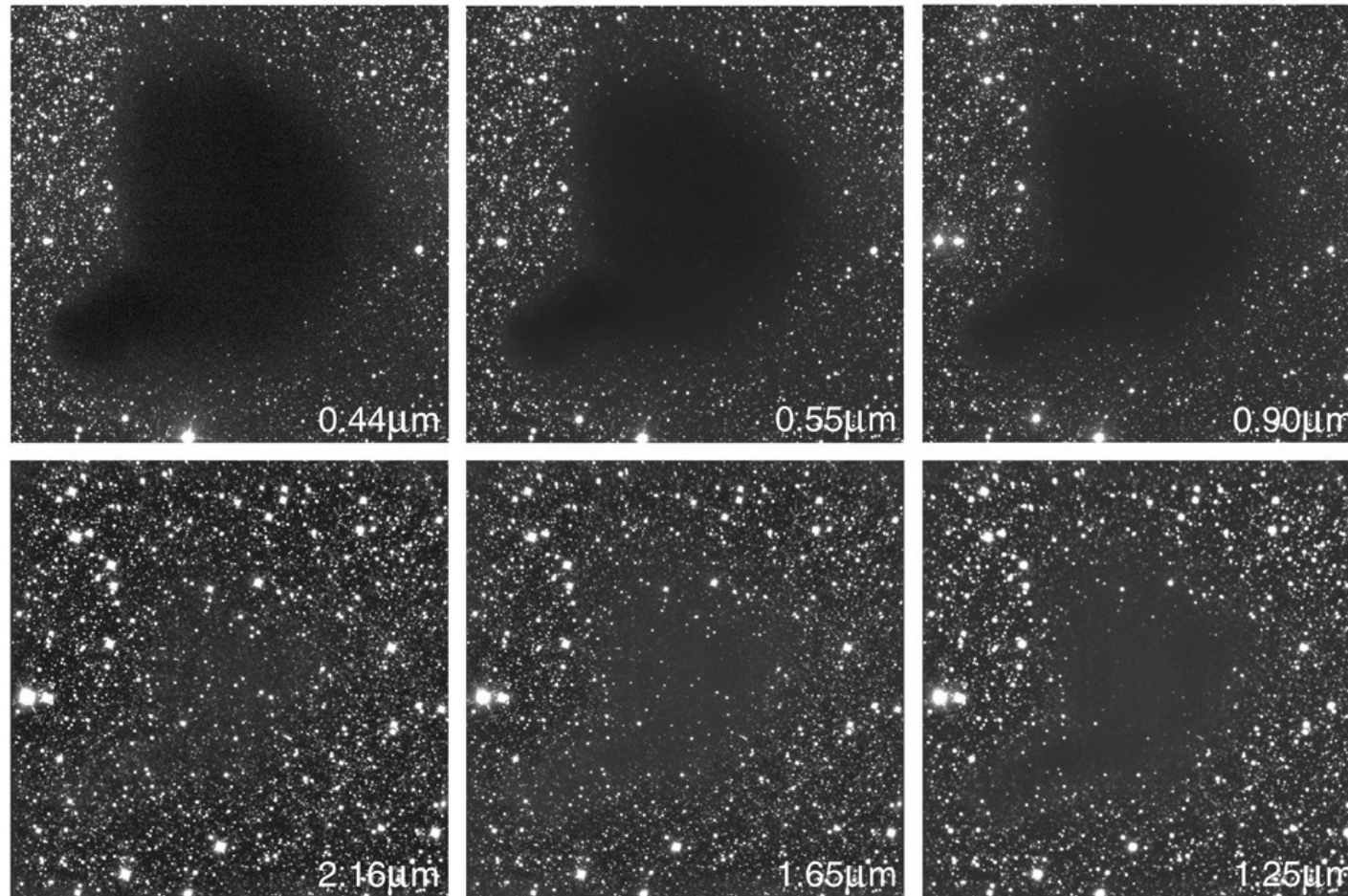
Interstellar Medium

- ISM is mostly confined to galactic disk
 - $\sim 10^9 M_{\odot}$
 - $n \sim 1 - 10 \text{ cm}^{-3}$
 - Mostly H (as HI, HII, or H₂)
- Radiation is not in equilibrium with gas
- Most of ISM mass is in clouds
- Denser clouds allow molecules to form
- Stars form in the densest clouds

How Does a Star Interact with ISM?

- Star formation:
 - Stars form from clouds of gas and dust in interstellar medium
 - Large clouds collapse under their own self gravity
 - Previous generations of stars enrich gas with metals
 - We've seen how stellar structure depends on metallicity, Z
- Star Death
 - Supernova return heavy elements to the ISM
 - Low mass stars lose mass via winds, with some chemical enrichment
 - Supernova shocks may be triggers for star formation

ISM



The Dark Cloud B68 at Different Wavelengths (NTT + SOFI)



Visible



Infrared

Visible vs. Infrared View of Pillar and Jets HH 901/902
Hubble Space Telescope • WFC3/UVIS/IR

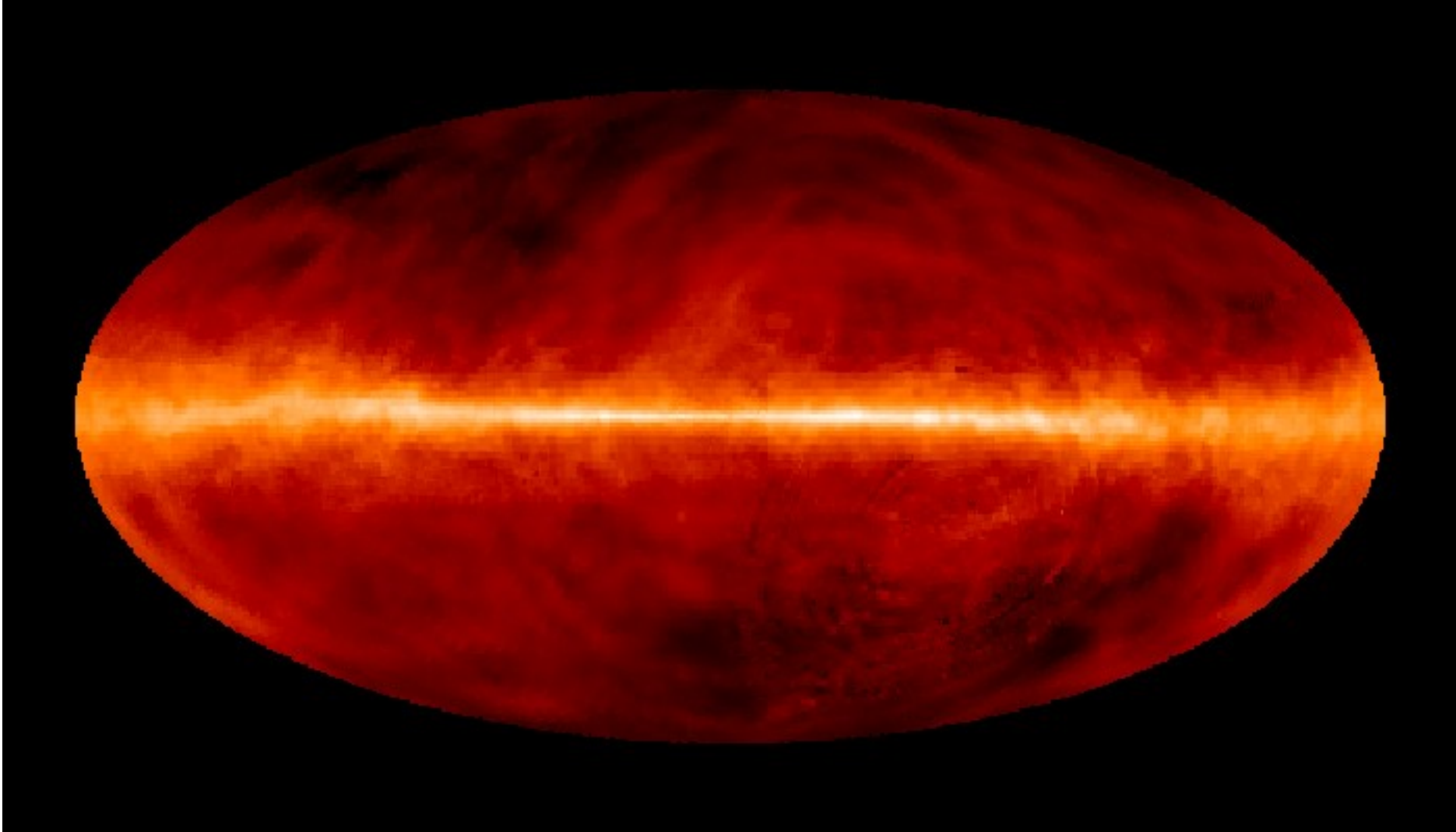
Pillars of Creation

NASA's Hubble Space Telescope made the Pillars of Creation famous with its first image in 1995, but revisited the scene in 2014 to reveal a sharper, wider view in visible light, shown above at left.

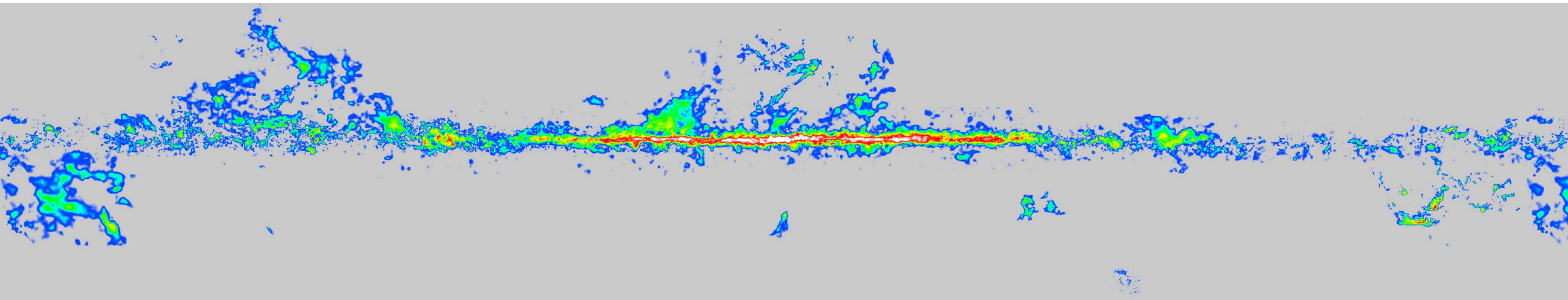
A new, near-infrared-light view from NASA's James Webb Space Telescope, at right, helps us peer through more of the dust in this star-forming region. The thick, dusty brown pillars are no longer as opaque and many more red stars that are still forming come into view



Neutral H Map

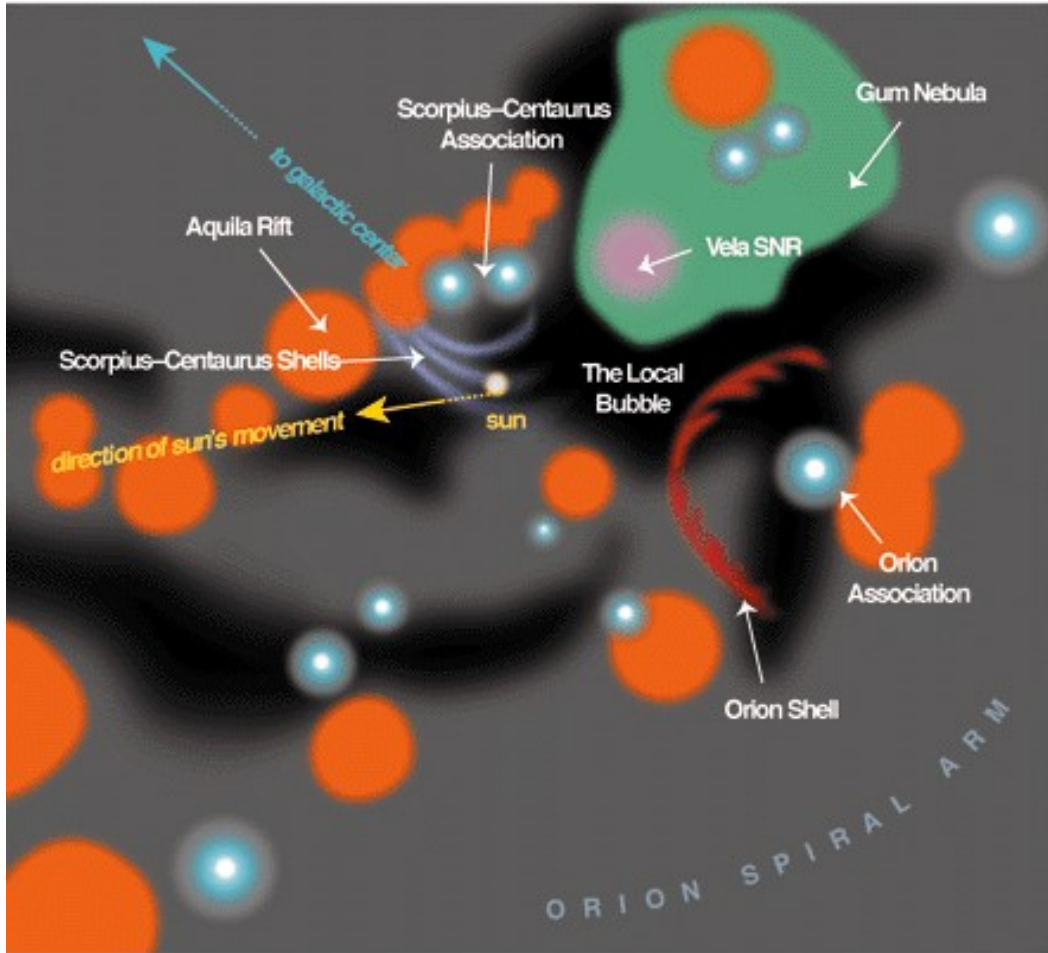


Molecular Gas Map



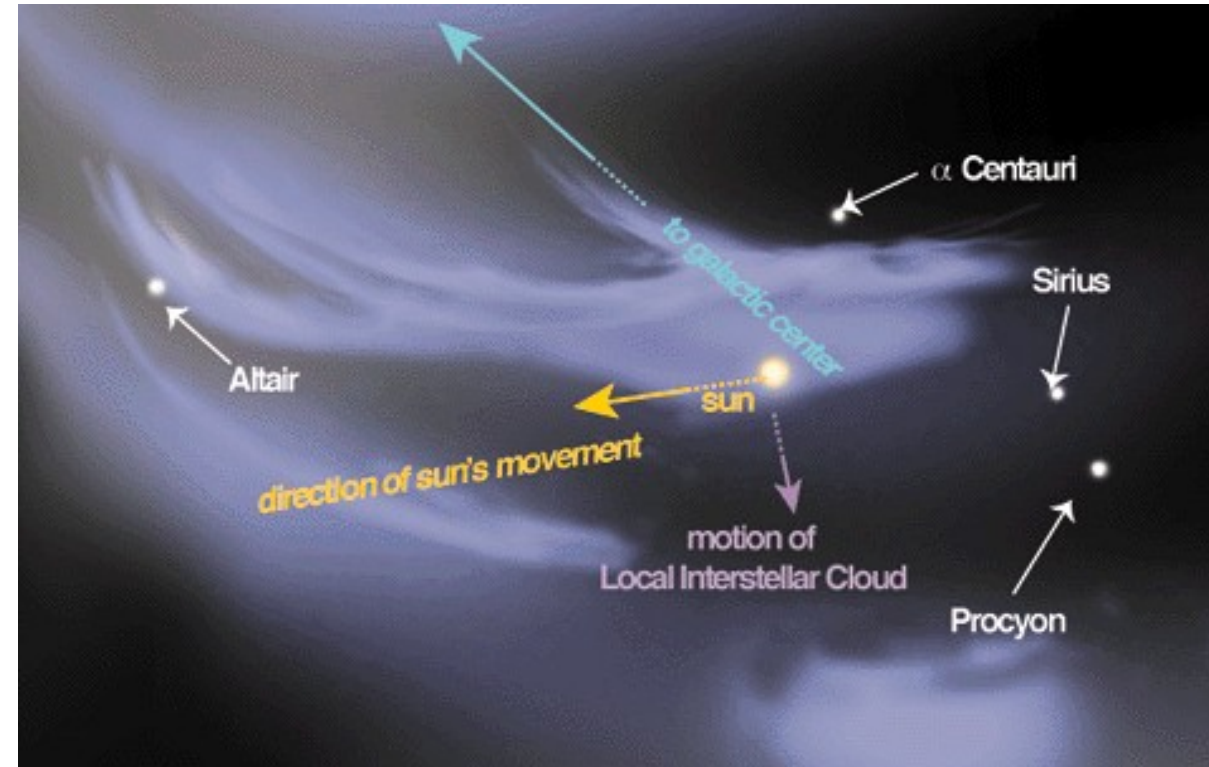
T. Dame ([CfA](#), [Harvard](#)) et al., Columbia 1.2-m Radio Telescopes
<http://antwrp.gsfc.nasa.gov/apod/ap970430.html>

Local Bubble



molecular clouds
 diffuse gas

Linda Huff ([American Scientist](http://antwrp.gsfc.nasa.gov/apod/ap020217.html)), [Priscilla Frisch \(U. Chicago\)](http://antwrp.gsfc.nasa.gov/apod/ap020217.html)
<http://antwrp.gsfc.nasa.gov/apod/ap020217.html>



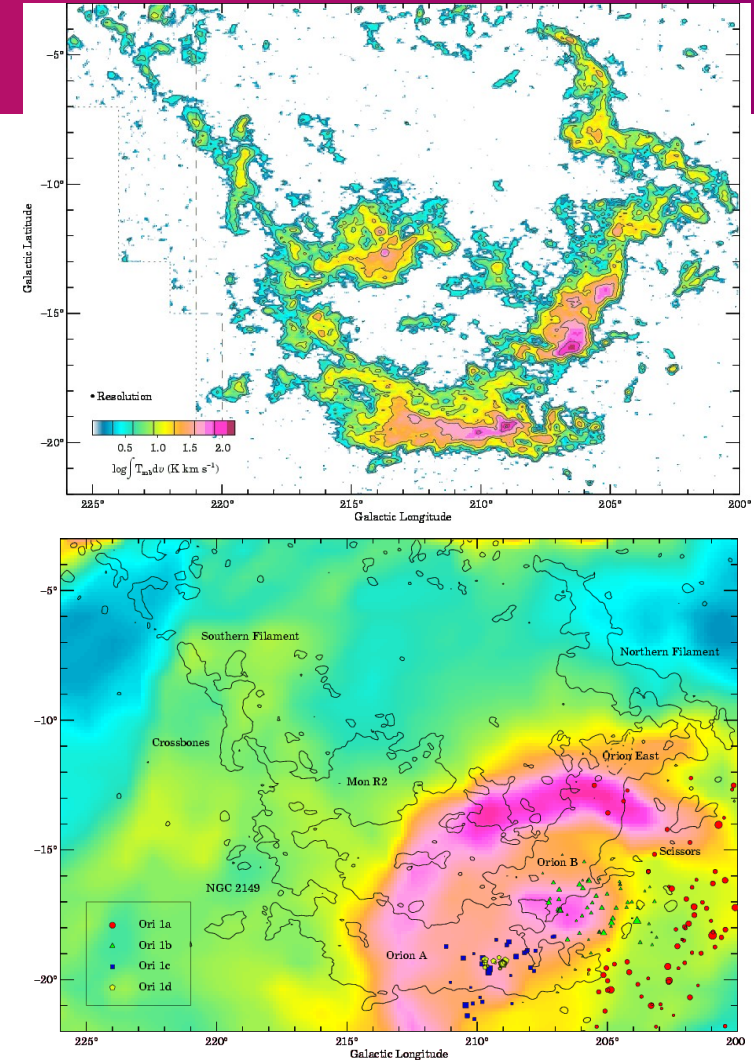
<http://antwrp.gsfc.nasa.gov/apod/ap020210.html>
 Linda Huff ([American Scientist](http://antwrp.gsfc.nasa.gov/apod/ap020210.html)), [Priscilla Frisch \(U. Chicago\)](http://antwrp.gsfc.nasa.gov/apod/ap020210.html)

A model of the local ISM. The Sun is believed to be moving through the **local interstellar cloud** just on the edge of the **local bubble**.

PI-

Star Formation

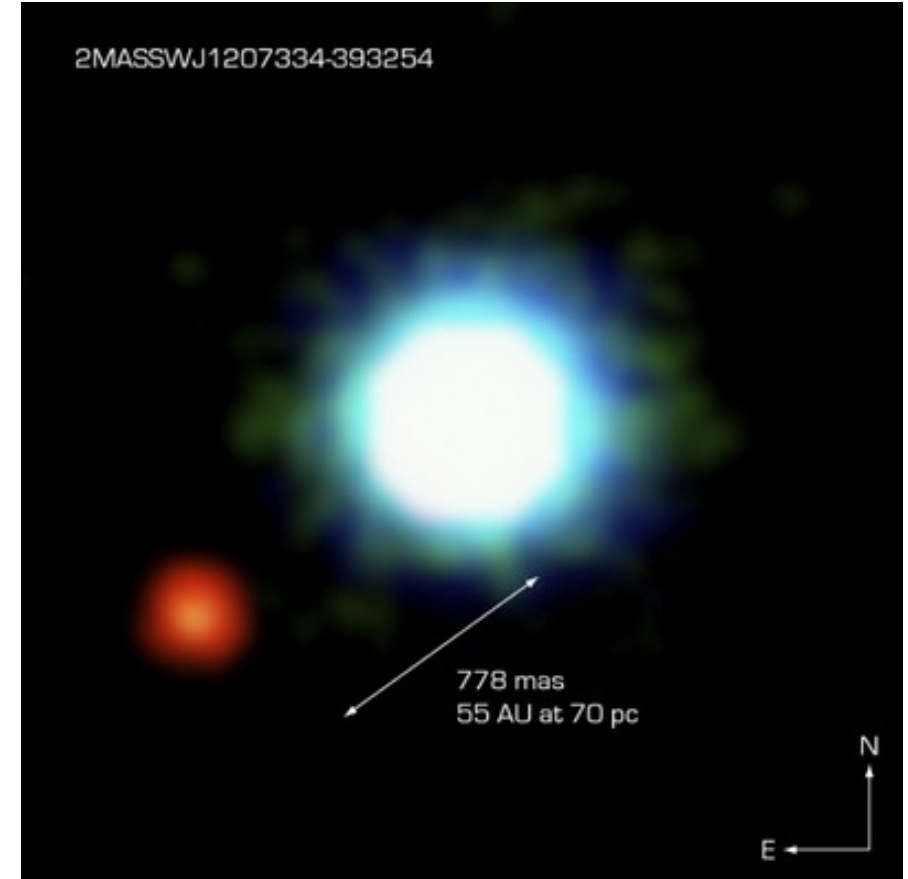
- Only the largest clouds in the ISM will collapse
- Fragmentation can occur as density increases and Jeans length shrinks



CO emission (top) and intensity of ionized gas (bottom) around Orion (B. A. Wilson - T. M. Dame - M. R. W. Mashedier - P. Thaddeus, 2005)

Brown Dwarfs

- Collapse of a cloud doesn't have to make just stars
 - Lower mass objects may also form this way
- Difference depends on whether H is ignited
- Brown dwarfs have $M < 0.08 M_{\odot}$
 - Still descend the Hayashi track
 - Contract and heat up (as do stars)
 - Reach degeneracy before crossing H ignition curve
- Planets formation is a different process



The Brown Dwarf 2M1207 and its Planetary Companion
(VLT/NACO)

ESO PR Photo 14a/05 (30 April 2005)



Brown Dwarfs

- Brown dwarfs form like stars, cool like planets
 - Can briefly burn ^2H
 - Luminosity vs. time is $\sim$ constant during ^2H burning
 - Cool after ^2H exhaustion
- Planets are $< 0.01 M_{\odot}$
 - No ^2H burning
 - Luminosity continually decreases

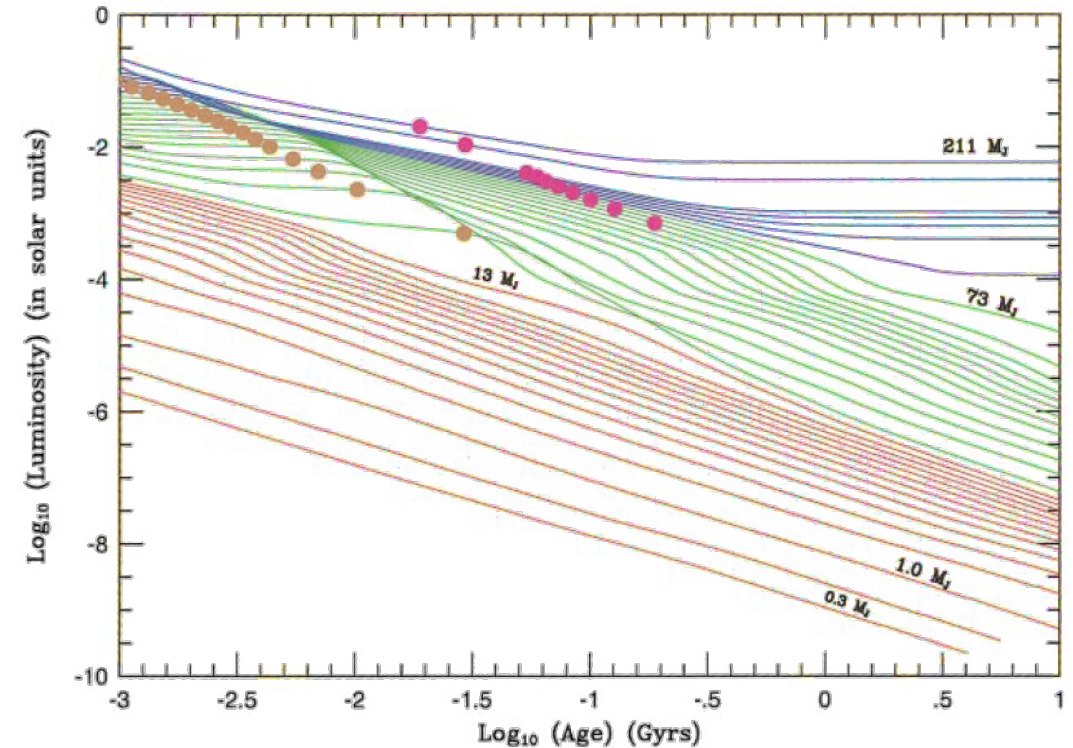
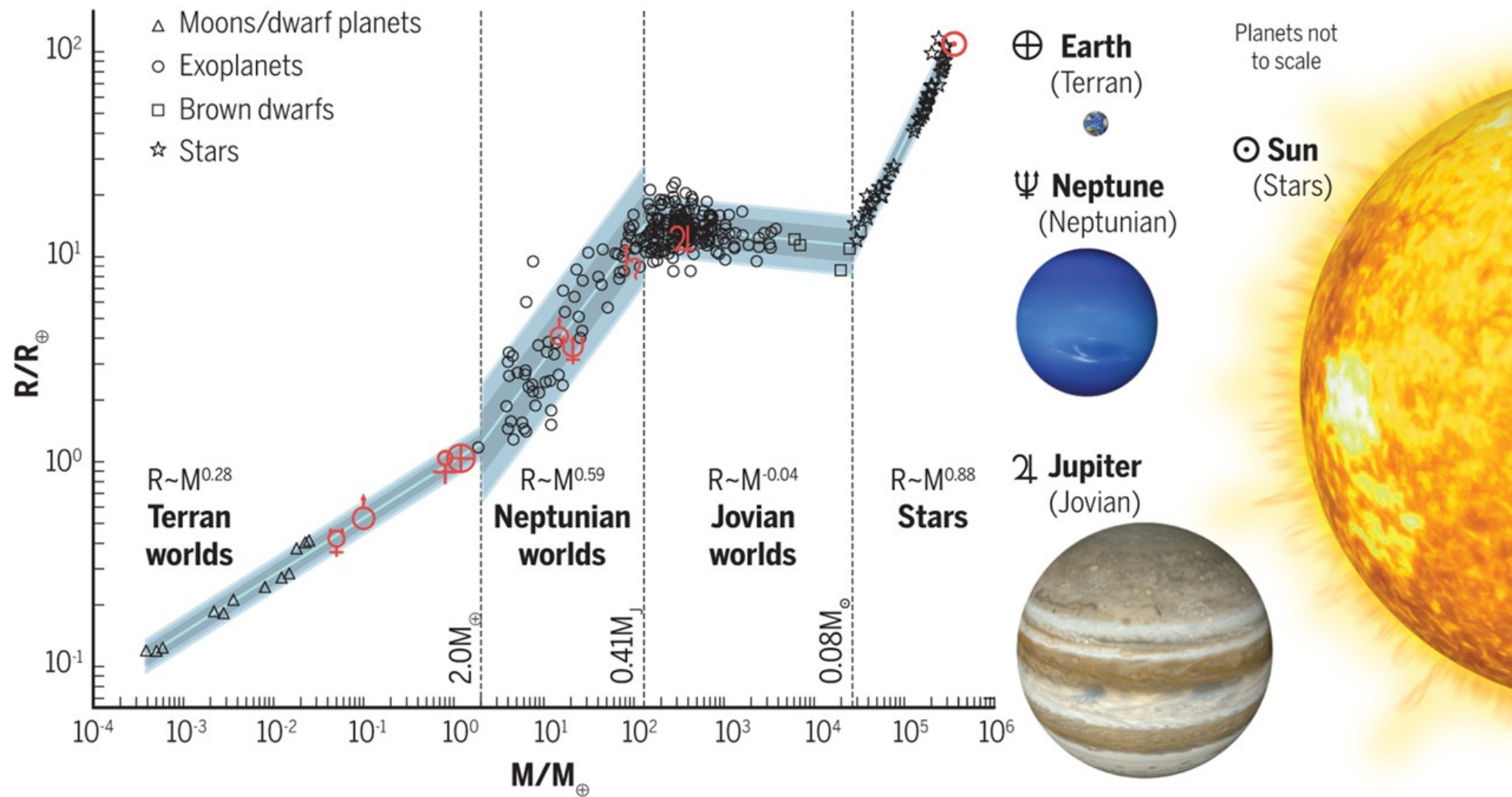


FIG. 1. Evolution of the luminosity (in $L_{\odot}$) of isolated solar-metallicity red dwarf stars and substellar-mass objects versus age (in years). The stars are shown in blue, those brown dwarfs above $13 M_J$ are shown in green, and brown dwarfs/giant planets equal to or below $13 M_J$ are shown in red. Though the color categories are based on deuterium or light hydrogen burning, they should be considered arbitrary *vis à vis* whether the object in question is a brown dwarf or a planet, sensibly distinguished on the basis of origin. The masses of the substellar objects/stars portrayed are 0.3, 0.5, 1.0, 2.0, 3.0, 4.0, 5.0, 6.0, 7.0, 8.0, 9.0, 10.0, 11.0, 12.0, 13.0, and $15.0 M_J$ and 0.02, 0.025, 0.03, 0.035, 0.04, 0.045, 0.05, 0.055, 0.06, 0.065, 0.07, 0.075, 0.08, 0.085, 0.09, 0.095, 0.1, 0.15, and $0.2 M_{\odot}$ ($=211 M_J$). For a given object, the gold dots mark when 50% of the deuterium has been burned and the magenta dots mark when 50% of the lithium has been burned. Note that the lithium sequence penetrates into the brown dwarf regime near $0.065 M_{\odot}$, below the HBMM. Figure based on Fig. 7 of Burrows *et al.*, 1997 [Color].

Brown Dwarfs

- Degeneracy is dominant pressure support in lowest mass stars and brown dwarfs
 - Radius increases with decreasing mass for complete degeneracy
 - Planets have more complex equations of state, with radius increasing with mass
 - Transition at some mass near Jupiter's mass
- Boundary between brown dwarf and planet is near Jupiter mass

Brown Dwarfs



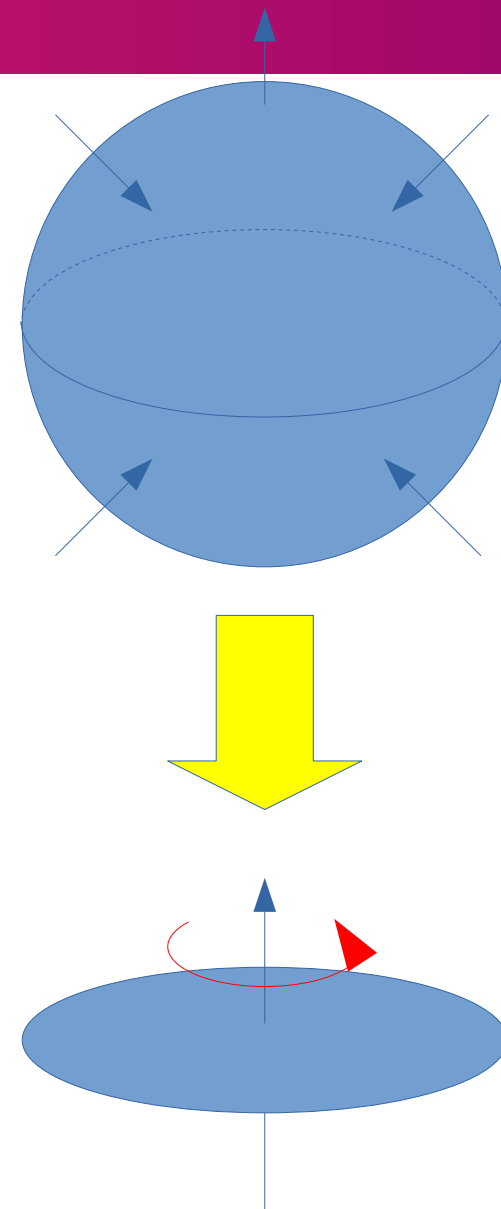
GRAPHIC: C. BICKEL/SCIENCE

<https://science.sciencemag.org/content/353/6300/644/tab-figures-data>

PHY521: Stars

How It Begins

- Solar nebula: gas between stars
- Solar nebula collapsed
 - Some gravitational PE radiated away, rest heated cloud
 - Angular momentum conservation → disk forms
- Sun forms at center
 - Eventually T high enough for fusion
 - H burning began
 - Energy produced in core matched energy radiated from surface—equilibrium
- Planets form in disk
 - Orbital directions reflect spinning disk



Star Formation in Orion



More than 3000 stars are in this image
amongst the gas and dust in the nebula

(NASA, ESA, M. Robberto (Space Telescope Science Institute/ESA)
and the Hubble Space Telescope Orion Treasury Project Team)



(Credit: Mouser Williams)

Jeans' Mass

- A “gaseous sphere” has two influences acting on it:
 - gravity pulling it together
 - kinetic energy of particles tending to cause it to fly apart.
- Static solution: critical mass such that mass lower than this limit will disperse and mass above this limit will collapse.
 - Jeans critical mass.
- For a given density, one can also consider a Jeans radius.

- Compare sound speed to free-fall collapse

- Collapse occurs if:

$$t_{\text{ff}} < t_{\text{sound}}$$
$$\frac{1}{\sqrt{G\rho}} < \frac{R}{c_s}$$
$$\frac{1}{\sqrt{G\rho}} < \frac{R}{\sqrt{P/\rho}}$$

- This gives the Jeans' length

Alternate Derivation

- Evaluating:

$$\frac{1}{\sqrt{G\rho}} \sim \frac{R}{c_s}$$

$$\frac{1}{\sqrt{G\rho}} \sim \frac{R}{\sqrt{P/\rho}}$$

- Jeans length:

$$R \sim \sqrt{\frac{P}{\rho}} \frac{1}{\sqrt{G\rho}} \sim \sqrt{\frac{kT}{G\mu m_u}} \frac{1}{\rho^{1/2}}$$

- Jeans mass:

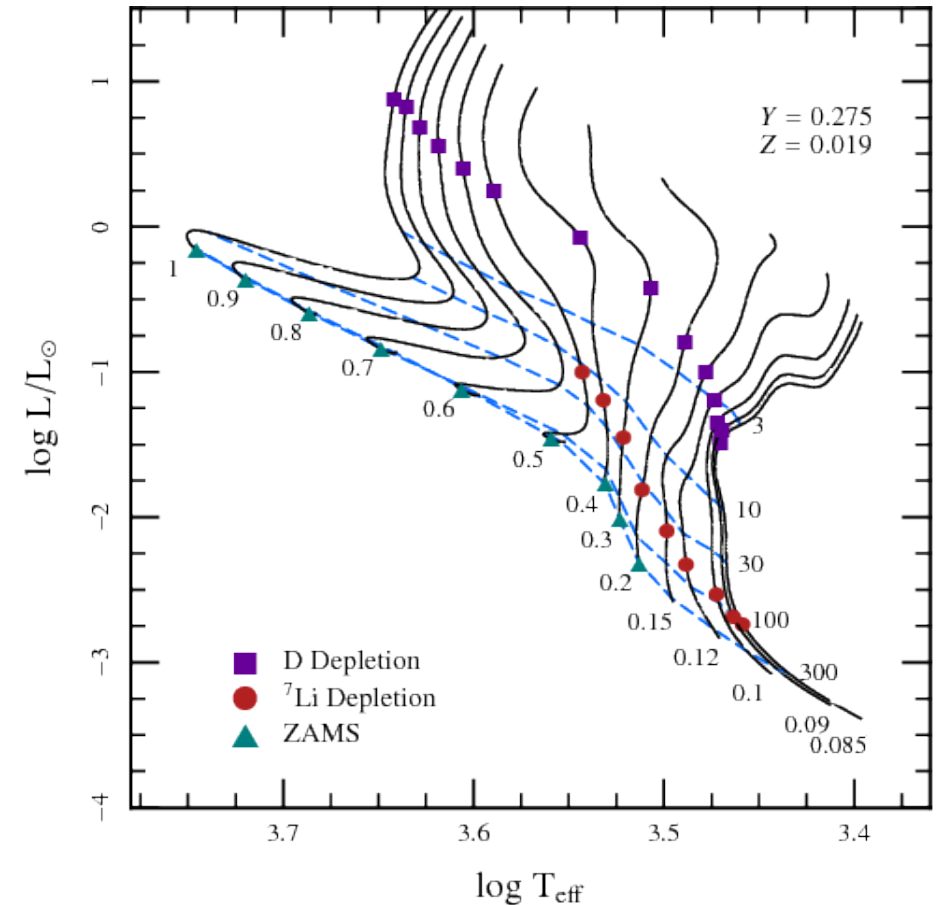
$$M_J \sim \frac{4}{3}\pi R_J^3 \rho$$

$$M_J \approx 10^2 \left(\frac{T}{1 \text{ K}} \right)^{3/2} \left(\frac{n}{1 \text{ cm}^{-3}} \right)^{-1/2} M_\odot$$

- Typical value $\sim 10^4 M_\odot$

Star Formation on the HR Diagram

- Hayashi track: boundary separating convecting hydrostatic stars (left) and those that have no means of effective energy transport (right)
- Move off Hayashi track when reactions kick in
- First part of pp-chain and conversion of ^{12}C to ^{14}N via CNO
 - Low mass stars never have this happen
- Once hot enough, full H burning commences



(Paxton et al. 2010)

Fig. 15.— Location in the Hertzsprung-Russell (H-R) diagram for $0.085M_{\odot} < M < 1M_{\odot}$ stars as they arrive at the main sequence for $Y = 0.275$ and $Z = 0.019$. The mass of the star is noted by the values at the bottom of the line. The dashed blue lines are isochrones for ages of 3, 10, 30, 100 and 300 Myr, as noted to the right. The purple squares (red circles) show where D (^7Li) is depleted by a factor of 100. The green triangles show the ZAMS.

Hayashi Track

- We can derive the Hayashi track in much the same way we did WD cooling
- Consider the star to be a polytrope (very good for fully-convective protostar)
 - Connect interior structure to surface
 - Start with:

$$K = \frac{G}{n+1} \left[\frac{4\pi}{\xi^{n+1} (-\theta'_n|_{\xi_1})^{n-1}} \right]^{1/n} M_{\star}^{1-1/n} R_{\star}^{-1+3/n}$$

- From polytrope EOS:

$$P = K \rho^{1+1/n}$$

- we get:

$$\log P_p = \left(1 - \frac{1}{n}\right) \log M_{\star} + \left(-1 + \frac{3}{n}\right) \log R_{\star} + \left(1 + \frac{1}{n}\right) \log \rho_p + \alpha$$

- ‘p’ subscript for photosphere
- α is a constant

Hayashi Track

- HSE gives:

$$\frac{dP}{d\tau} = +\frac{g_{\star}}{\kappa}$$

- Integrating from the surface to the photosphere:

$$0 - P_p = \frac{g_{\star}}{\kappa} = -\frac{GM_{\star}}{\kappa R_{\star}^2}$$

- Take opacity as powerlaw:

$$\kappa = \kappa_0 \rho_p T_{\text{eff}}^{\nu}$$

- Together:

$$\log P_p = \log M_{\star} - 2 \log R_{\star} - \log \rho_p - \nu \log T_{\text{eff}} + \beta$$

Hayashi Track

- Ideal gas law:

$$\log P_p = \log \rho_p + \log T_{\text{eff}} + \gamma$$

- Blackbody relation:

$$\log L_{\star} = 2 \log R_{\star} + 4 \log T_{\text{eff}} + \delta$$

- 4 equations + 4 unknowns

- Assume H- opacity: $\nu = 4$

- Solution:

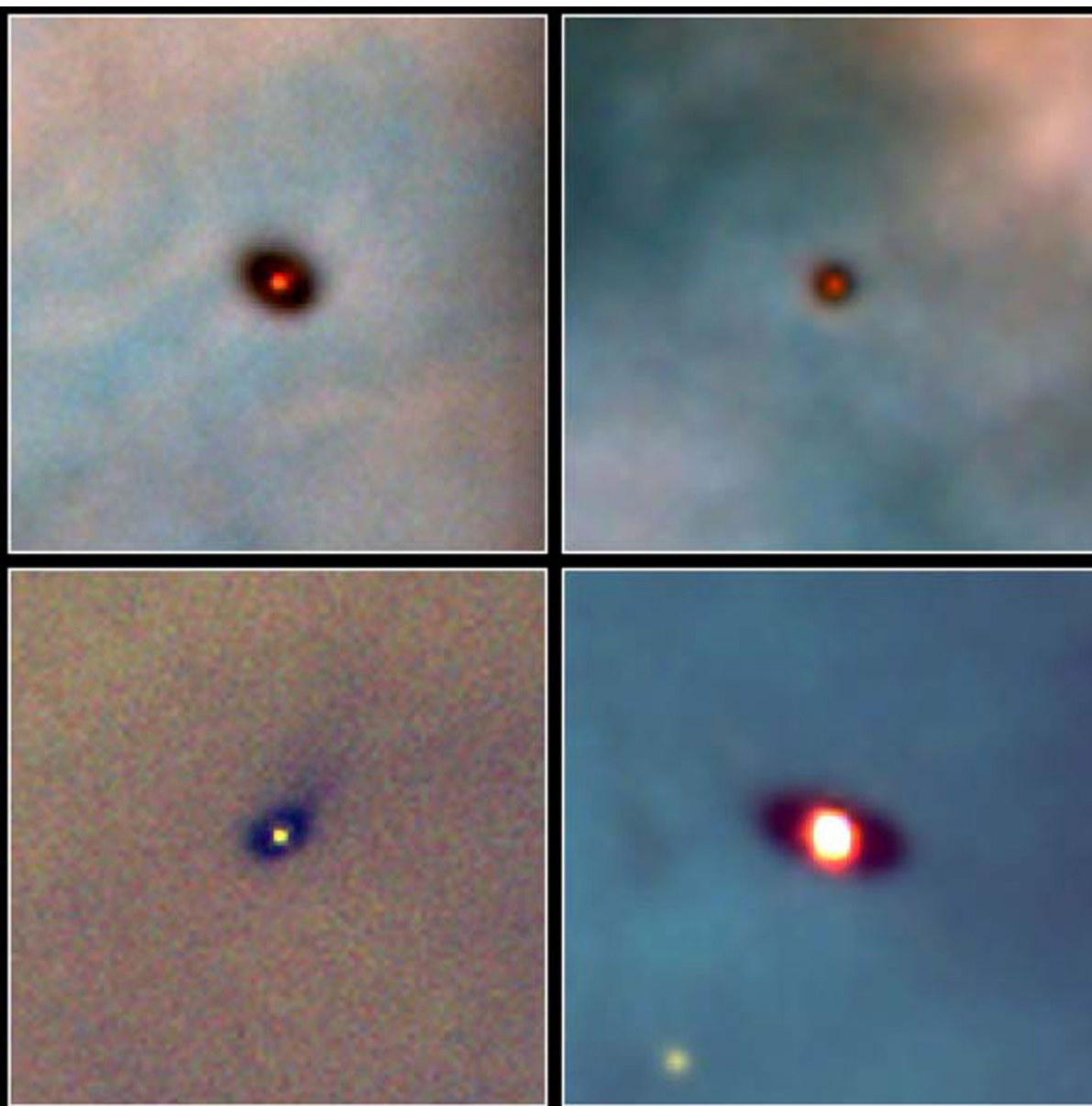
$$\log L_{\star} = -4 \log M_{\star} + 20 \log T_{\text{eff}} + c$$

- This is a really steep line in the HR diagram



Protoplanetary Disks in the Orion Nebula
Hubble Space Telescope • WFPC2

NASA, J. Bally (University of Colorado), H. Throop (SWRI), and C.R. O'Dell (Vanderbilt University)
STScI-PRC01-13



Protoplanetary Disks Orion Nebula

HST • WFPC2

PRC95-45b • ST ScI OPO • November 20, 1995

M. J. McCaughrean (MPIA), C. R. O'Dell (Rice University), NASA

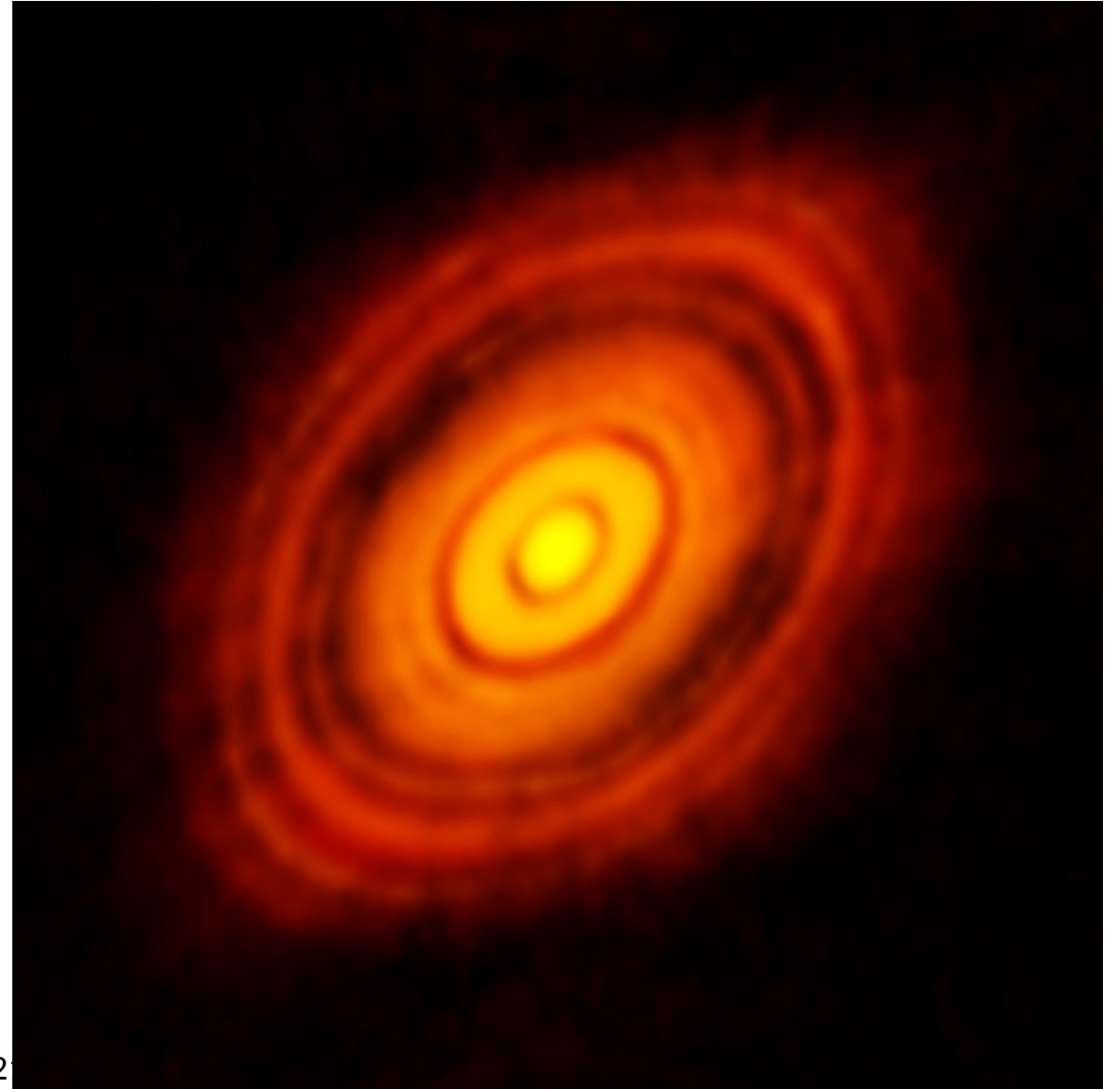


<http://antwrp.gsfc.nasa.gov/apod/ap091222.html>

Credit: NASA, ESA, M. Robberto (STScI/ESA), the HST Orion Treasury Project Team, & L. Ricci (ESO)

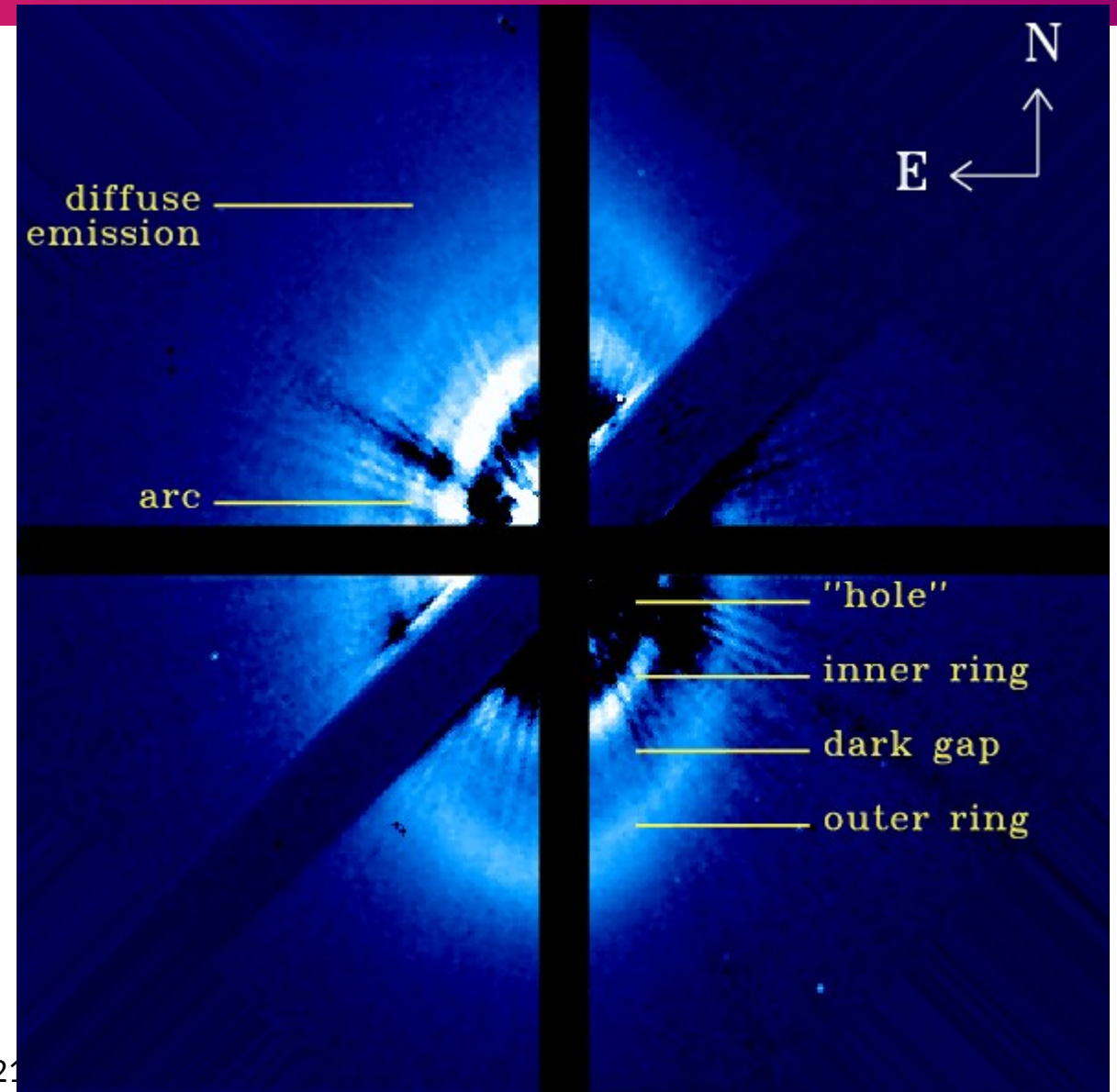
Young Disk

- HL Tau: < 1 Myr; 140 pc



Slightly Older Disk

- HD 141569; 5 Myr, 98 pc



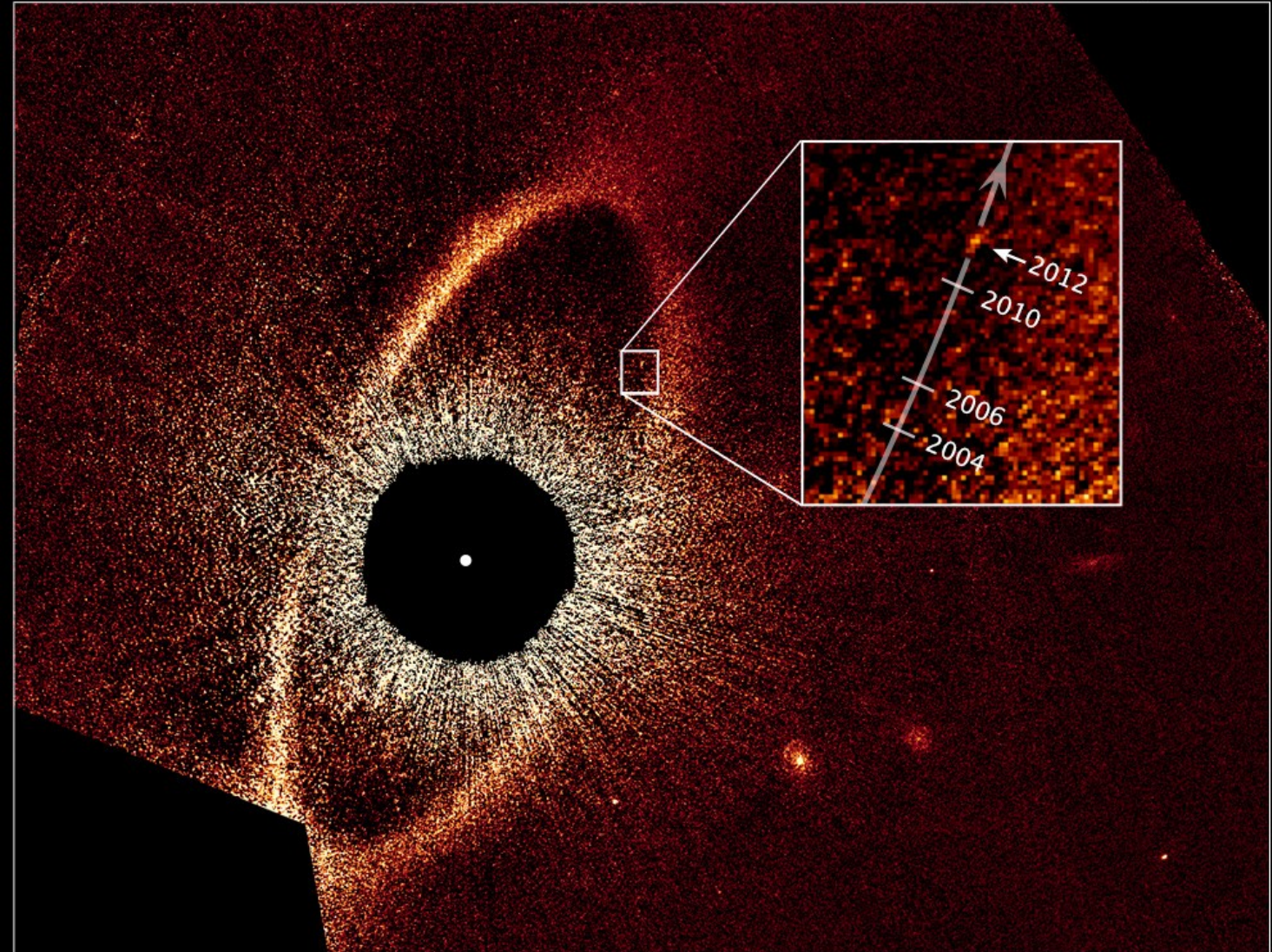
Debris Disk

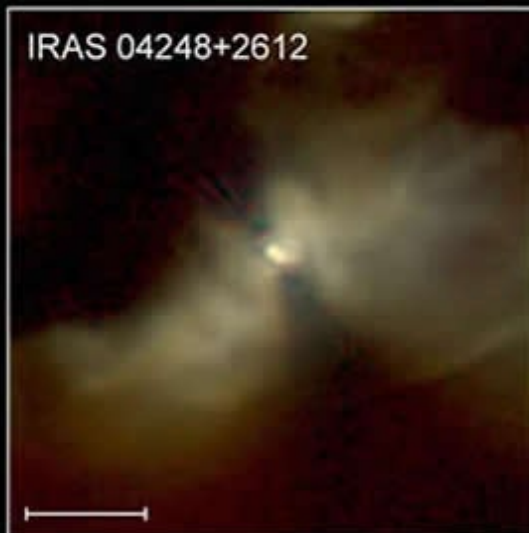
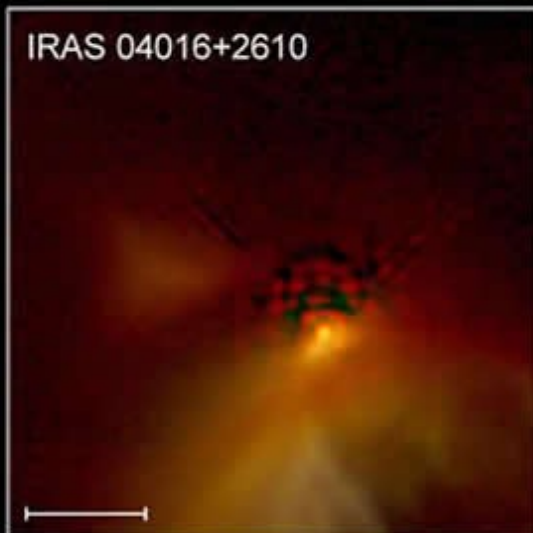
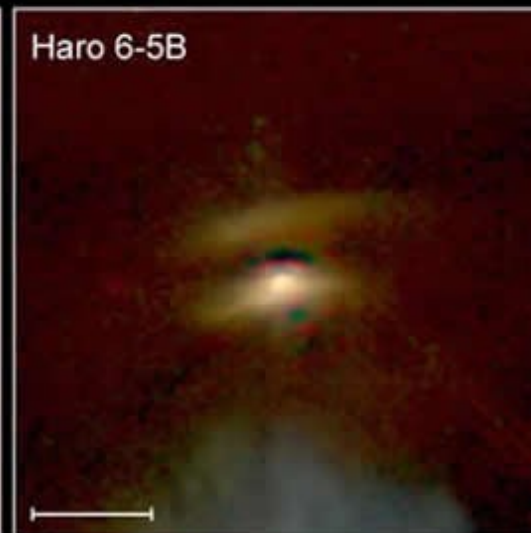
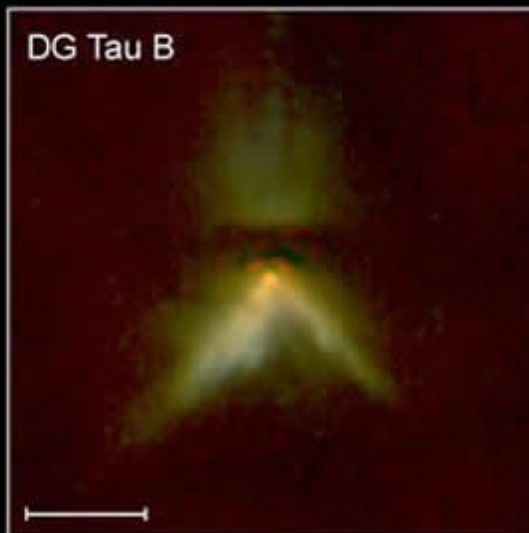
- Formalhaut 440 Myr, 7.6 pc

At this stage, a planet may be actively shaping the debris

Fomalhaut System

Hubble Space Telescope • STIS





Young Stellar Disks in Infrared
Hubble Space Telescope • NICMOS

Initial Mass Function

- Number of stars formed at a given time in a volume w/ $M \in [M, M + dM]$:

$$dN = \Phi(M)dM$$

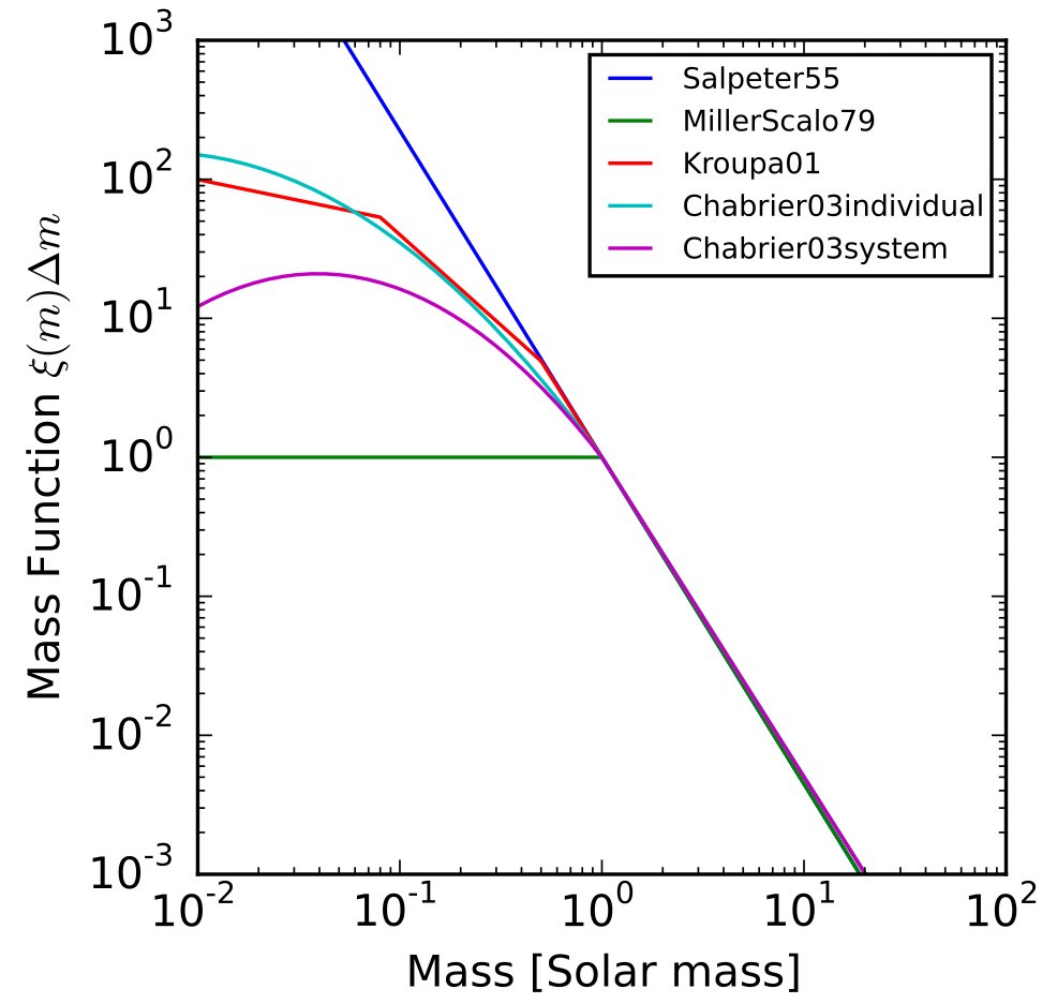
- $\phi(M)$ Is the “birth function”
- Salpeter: $\phi(M) \sim M^{-2.35}$

- Initial mass function:

$$MdN \equiv \xi(M)dM = \Phi(M)MdM$$

$$\xi(M) \propto \left(\frac{M}{M_{\odot}} \right)^{-1.35}$$

- Empirically obtained
- Low end is uncertain



Mass Return to ISM

- Consider a generation of stars at a given time

- Mass initially locked up in stars:

$$\xi = \int_{M_{\min}}^{M_{\max}} \xi(M) dM$$

- Stars lose mass as they evolve
 - $\mathcal{R}$ is fraction of initial mass ejected by star
 - $\mathcal{R}$ depends on mass of star
- Mass returned to ISM:

$$\eta = \int_{M_{\min}}^{M_{\max}} \xi(M) \mathcal{R}(M) dM$$

- What is $\mathcal{R}$?

$$\mathcal{R} = \begin{cases} 1 & M > M_{\text{SN}} \\ \frac{M - M_{\text{WD}}}{M} & M_{\text{MS}} < M < M_{\text{SN}} \\ 0 & M < M_{\text{MS}} \end{cases}$$

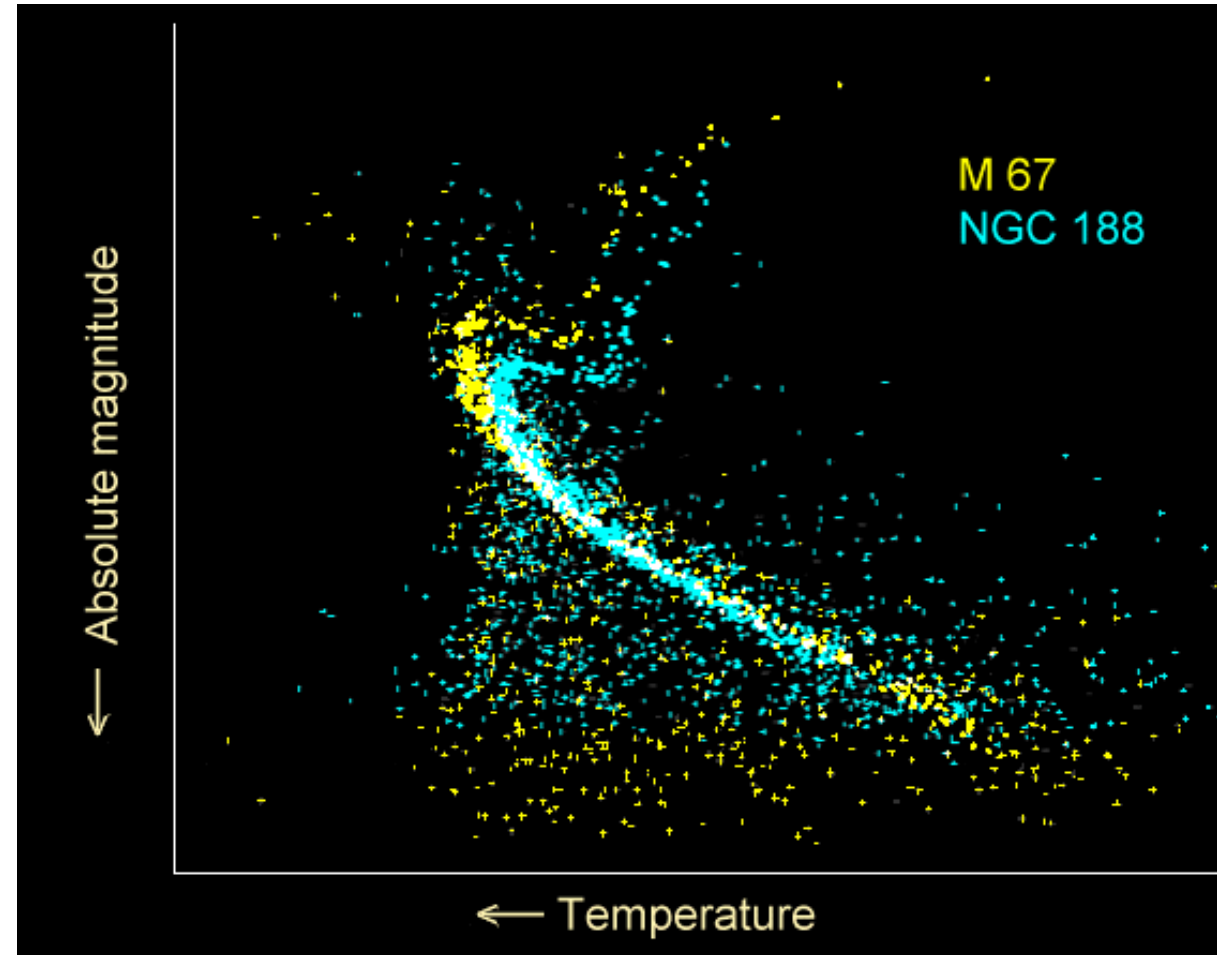
- Stars that supernova return essentially everything
- Stars that have evolved off the main sequence return anything above the WD mass
- Taking into account different mass losses, we find $\eta/\xi \sim 1/3$

What Fraction of Stars are WDs?

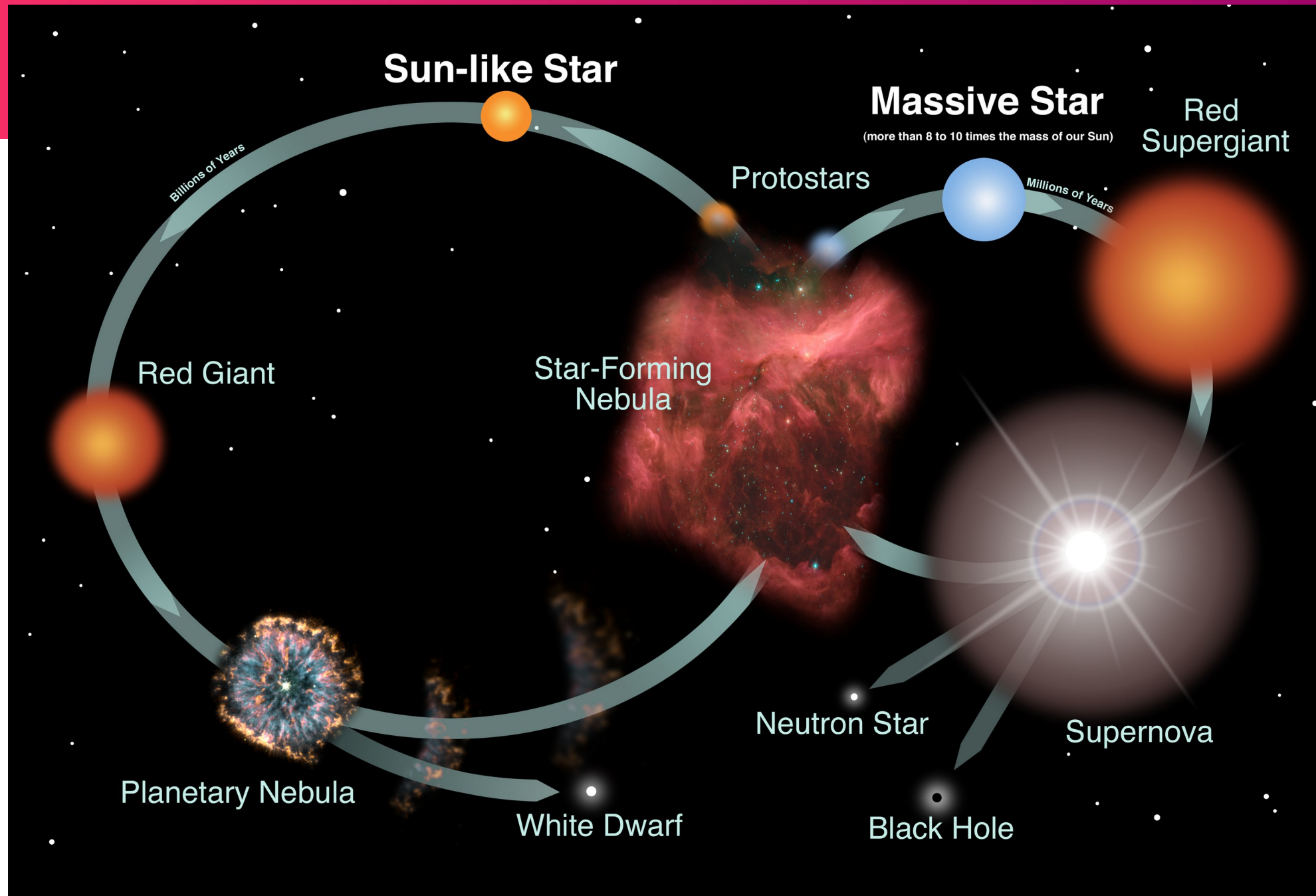
- Consider stars formed together in a cluster
- Main-sequence turnoff tells us which stars are still on the main sequence
- Fraction of stars that are WDs / MS:

$$\frac{N_{\text{WD}}}{N_{\text{MS}}} = \frac{\int_{M_{\text{tp}}}^{M_{\text{SN}}} \Phi(M) dM}{\int_{M_{\text{min}}}^{M_{\text{tp}}} \Phi(M) dM}$$

- Typically a few %



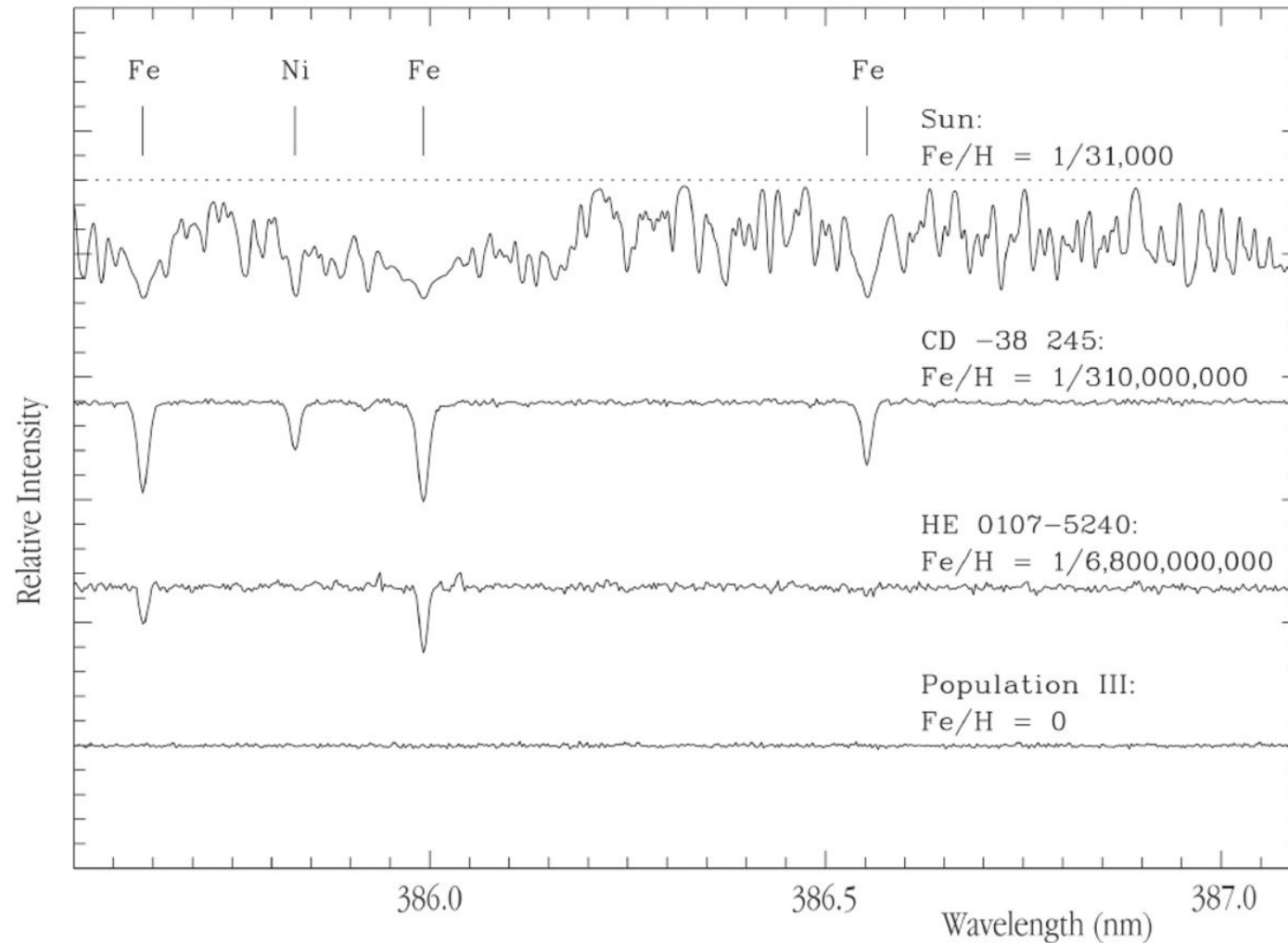
(Worldtraveller
/Wikipedia)



Stellar Life Cycle

- First generation of stars were zero metallicity (population III)
- Next generation were metal poor (population II)
- Latest generation is metal rich (population I)
- Universe is 13.6 billion years old
- Sun formed only recently
 - Enriched by metals produced in previous generation of stars ($Z \sim 0.02$)
- Newest stars have $Z \sim 0.04$
- Lowest mass stars never evolve – lock up gas
 - Eventually these will dominate

Stellar Populations



Spectra of Stars with Different Metal Content

Stellar Life Cycle

- Evolution in our Galaxy
 - Mass of free gas decreases – fewer nebula / clouds
 - Galactic luminosity decreases (fewer massive stars)
 - Composition is enriched by metals
- Massive stars provide most of the chemical enrichment to the galaxy
 - Most enrichment happened early

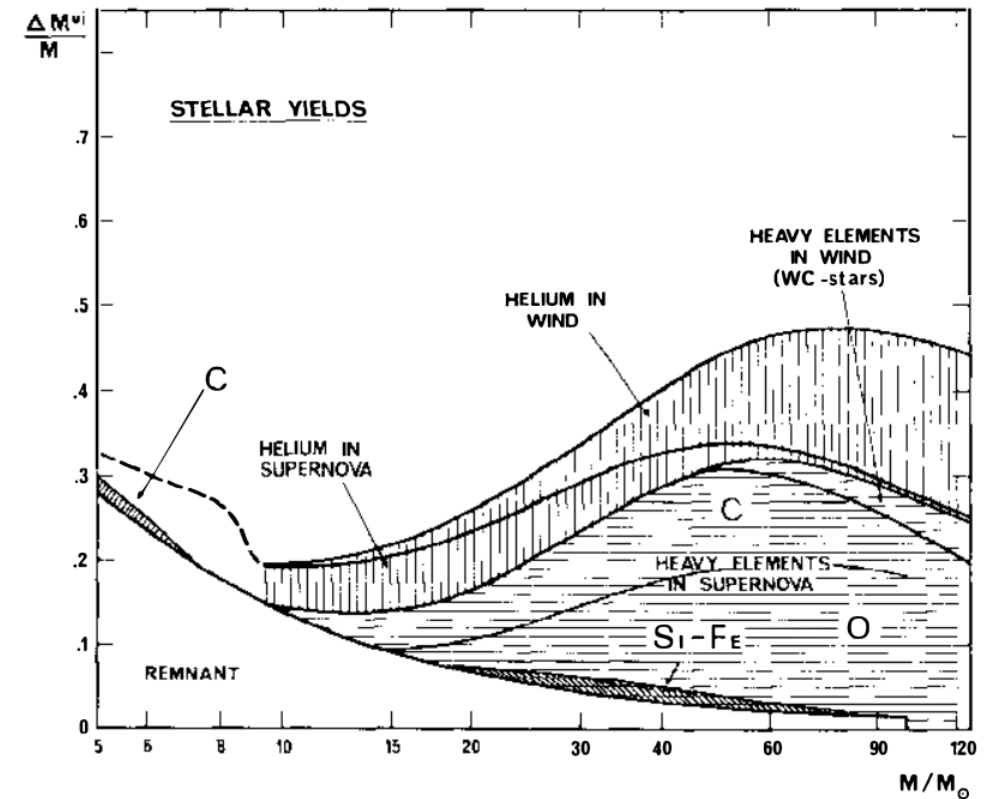


Figure 6 Mass fraction of new helium and new heavy elements ejected as a function of the initial stellar mass. Contributions from stellar winds of OB stars, supergiants, and WR stars are totaled and distinguished from the contribution from supernova explosions. The distribution of heavy elements in the supernova ejecta is based on the classic value for the $^{12}\text{C}(\alpha, \gamma)^{16}\text{O}$ reaction rate (see text). The contribution to ^{12}C by intermediate-mass stars is derived from Renzini & Voli (214) and is limited to their case A.

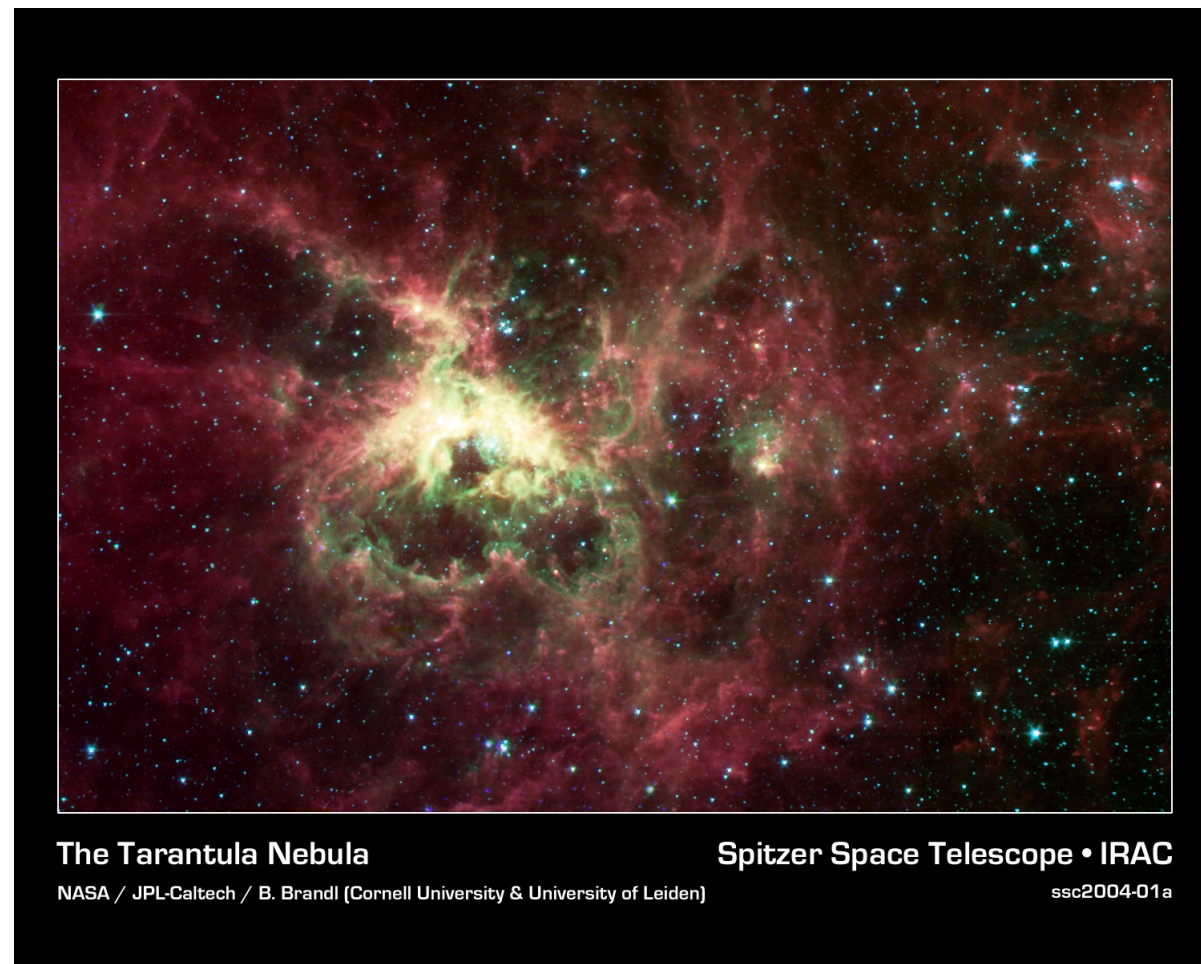
(Chiosi & Marder 1986)

Outstanding Questions

- Do we need a trigger to initiate the collapse?
 - Shock wave from nearby supernova?
 - Stellar wind from young star?
- How much of the mass of the cloud winds up in stars?
- What is the initial mass function of stars?
- OB associations: unbound stars from a bound cloud—how?
- Do high and low mass stars form in the same place?
 - High mass stars may form in bursts

HII Regions

- Hot stars (O, early B) have significant flux in UV
 - Able to ionize H
 - H II region: region surrounding star where ionization takes place—extends out to the point where ionization and recombination balance

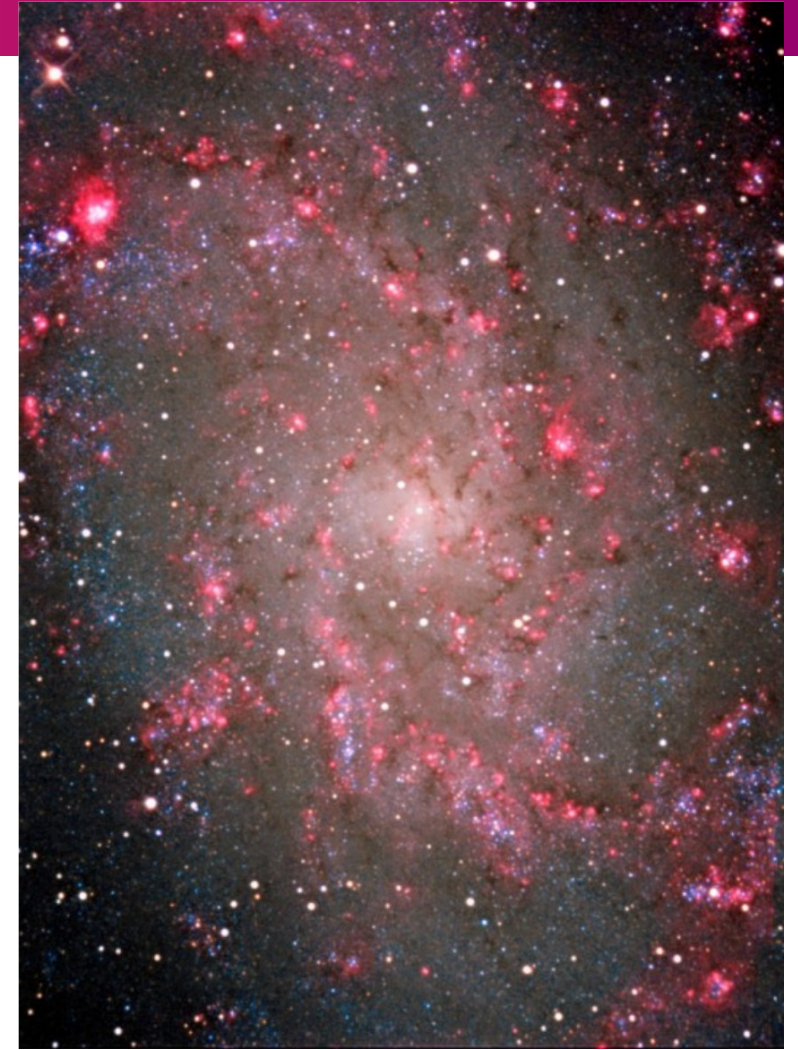


HII Regions



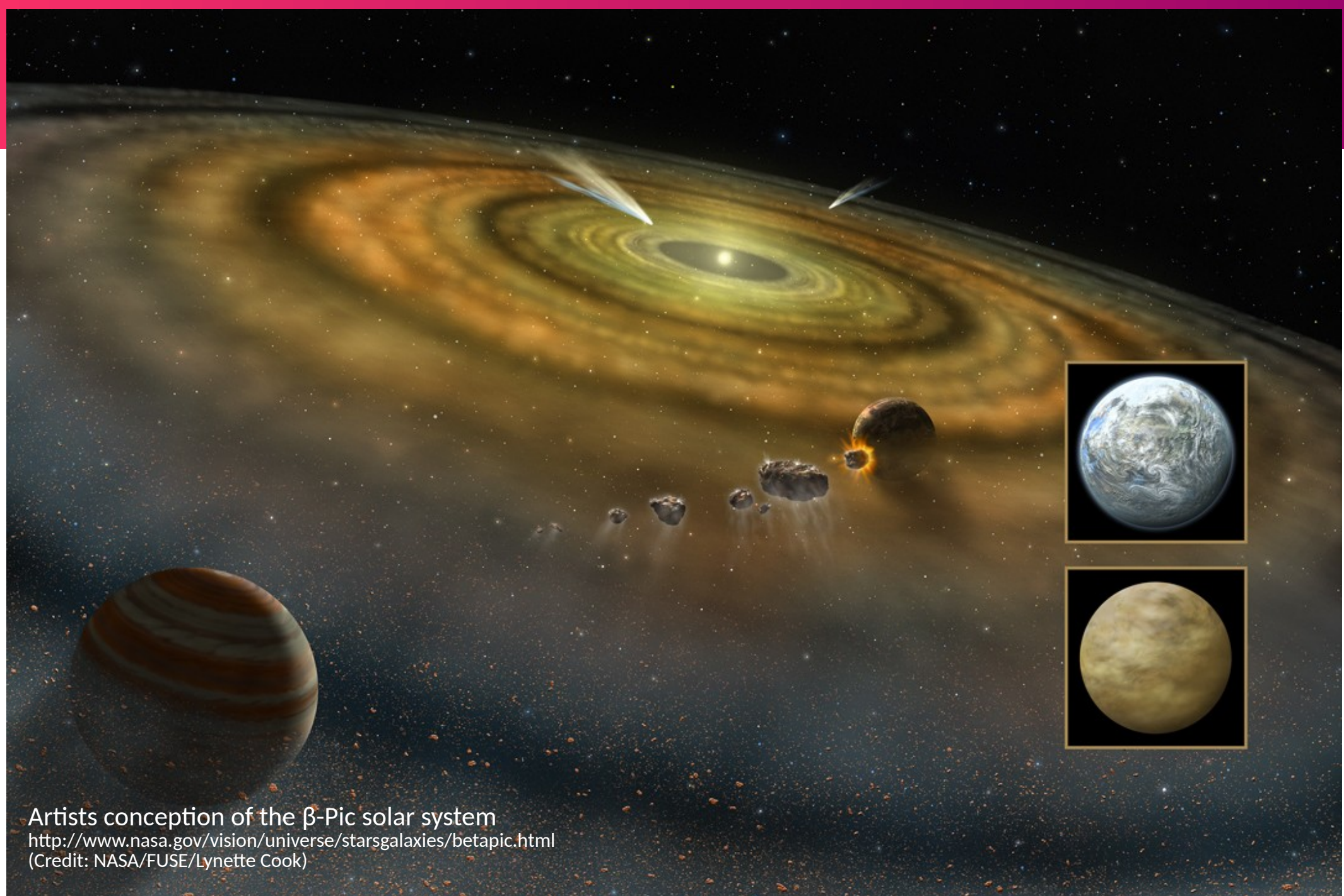
H II regions (red spots) in M51

(NASA, ESA, S. Beckwith (STScI), and The Hubble Heritage Team (STScI/AURA))



M33

(Chris Schur)



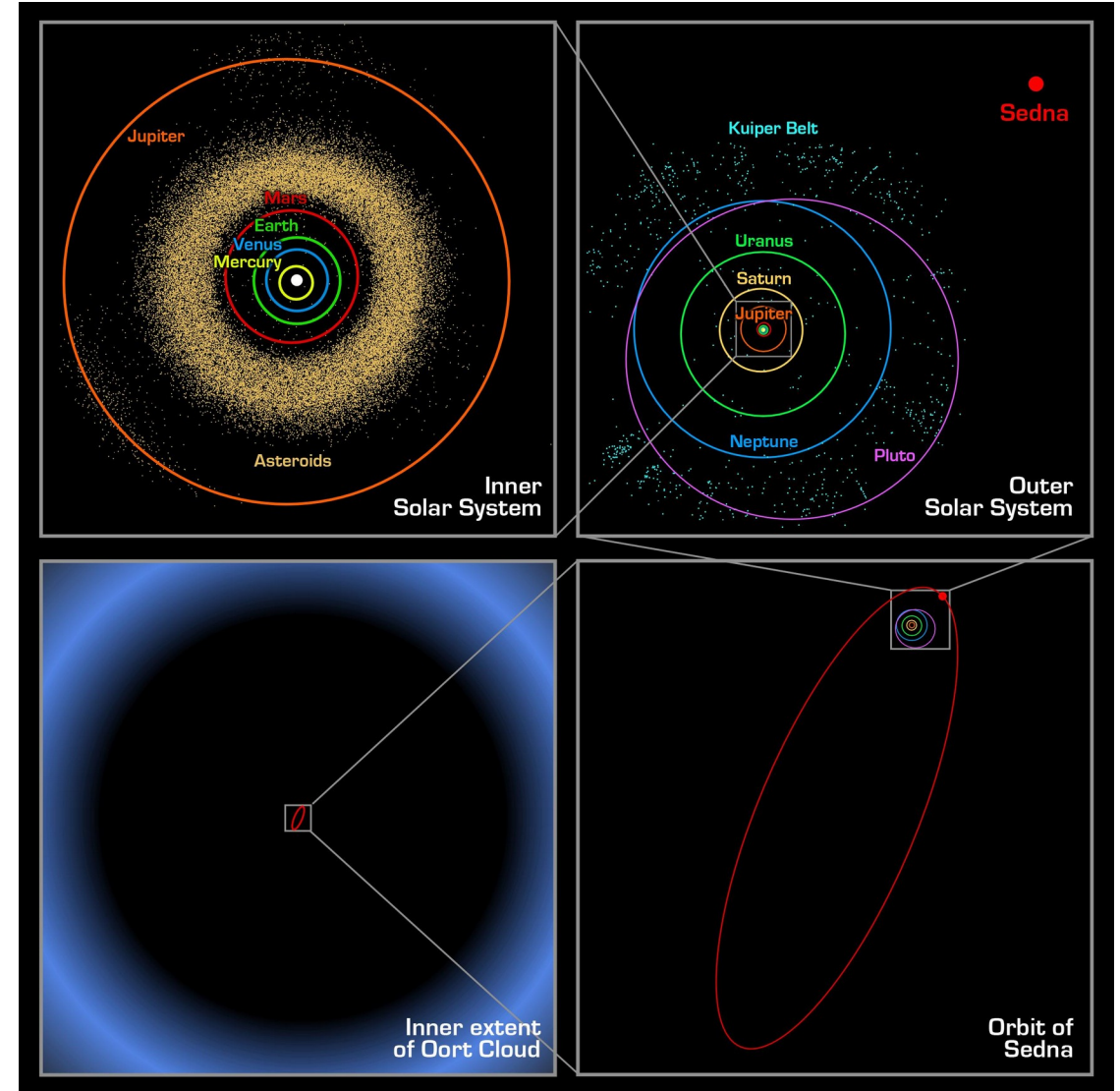
Artists conception of the β -Pic solar system

<http://www.nasa.gov/vision/universe/starsgalaxies/betapic.html>

(Credit: NASA/FUSE/Lynette Cook)

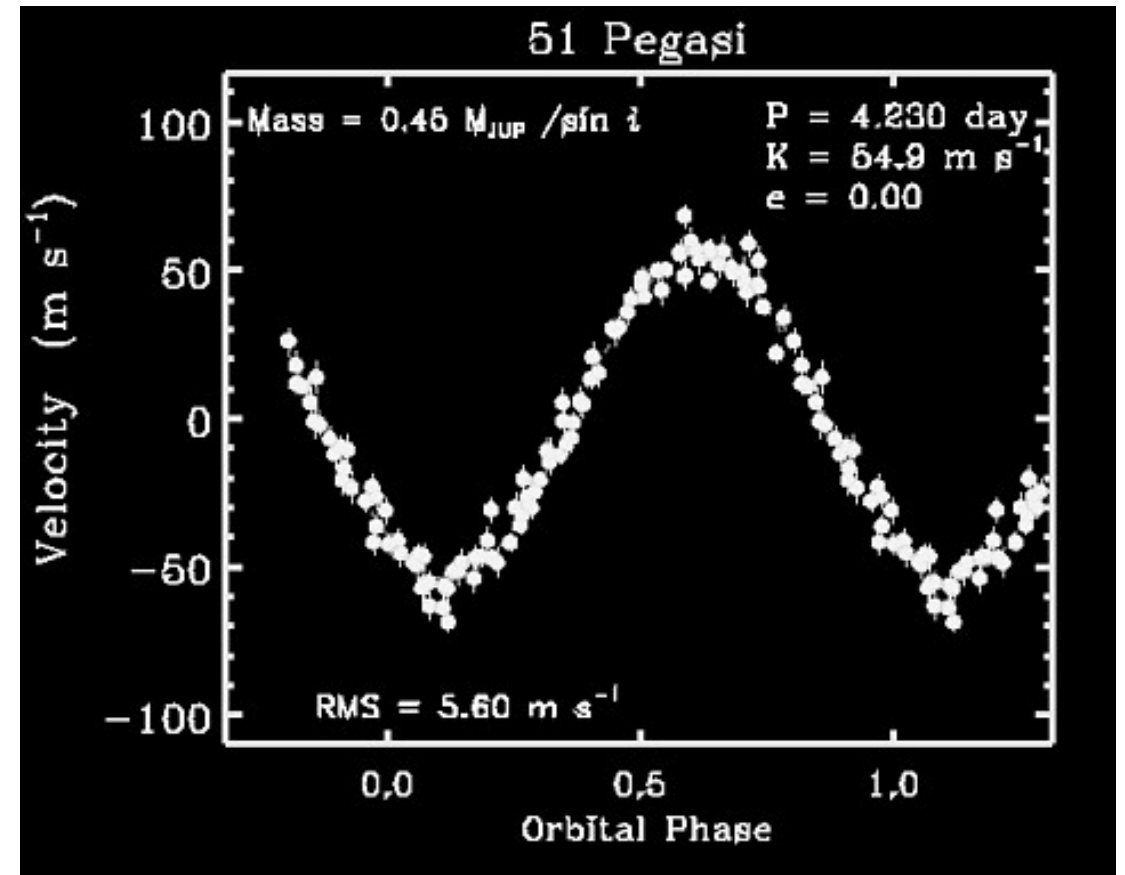
Solar System 101

- Organized motions
- Two types of planets
- Asteroids and comets
- Exceptions...



First Detection

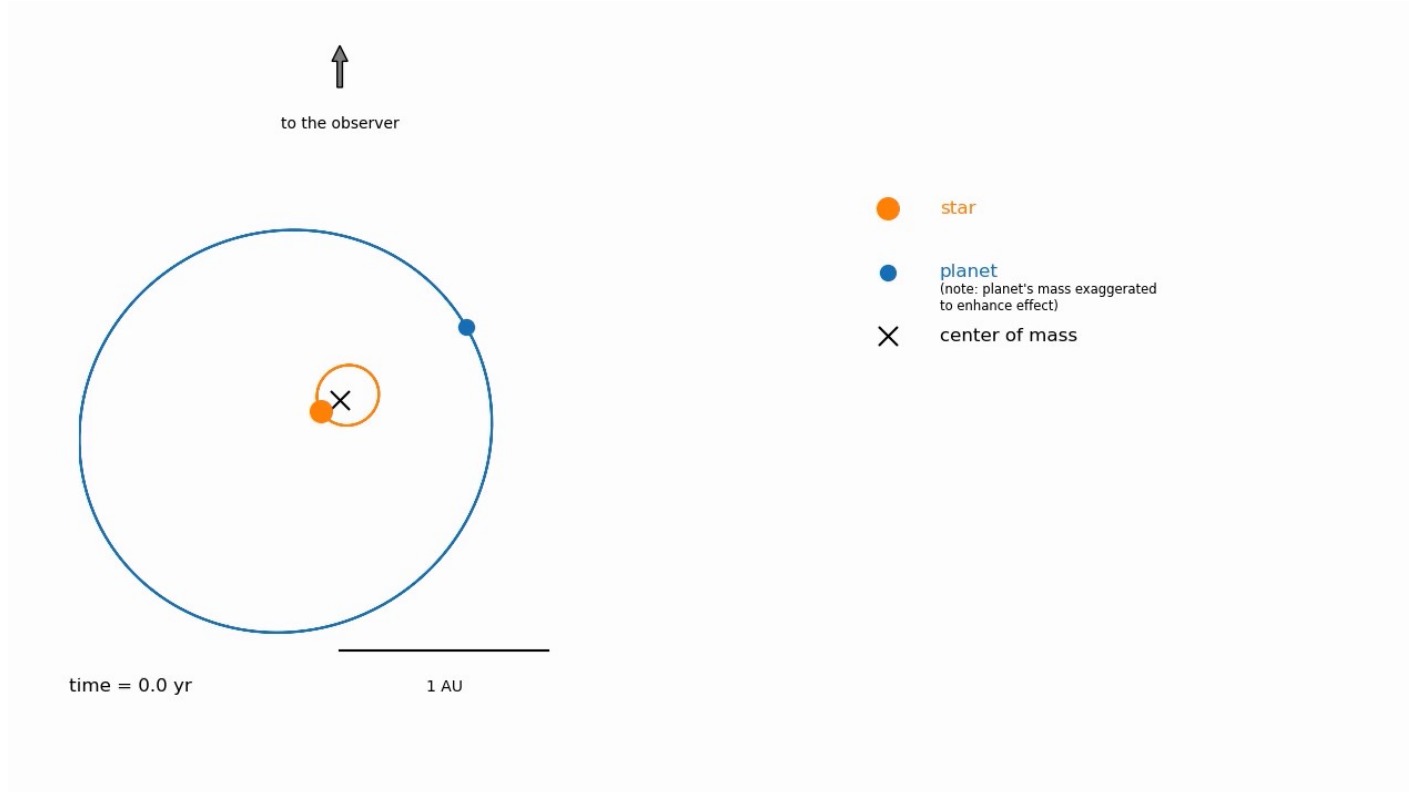
- 51 Pegasi b
 - Discovered via radial velocity technique
 - Note the period!



Radial velocity curve for 51 Pegasi

(from <http://www.extrasolar.net/planettour.asp?PlanetID=1>)

Radial Velocity Method



- Typical velocity ~ 10 s m/s
- Mass of star estimated via spectra
- Determines:
 - Period / semi-major axis
 - Eccentricity
 - $M \sin I$
 - Number of planets
- Biased toward massive planets close to star

Radial Velocity Method

- Multiple planets

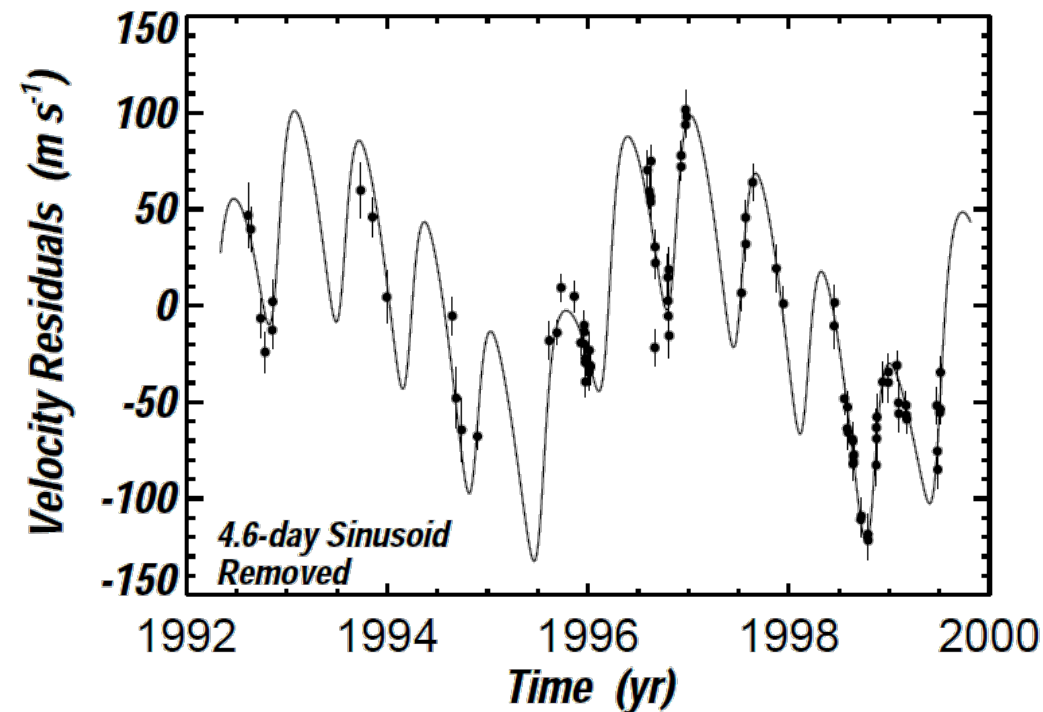
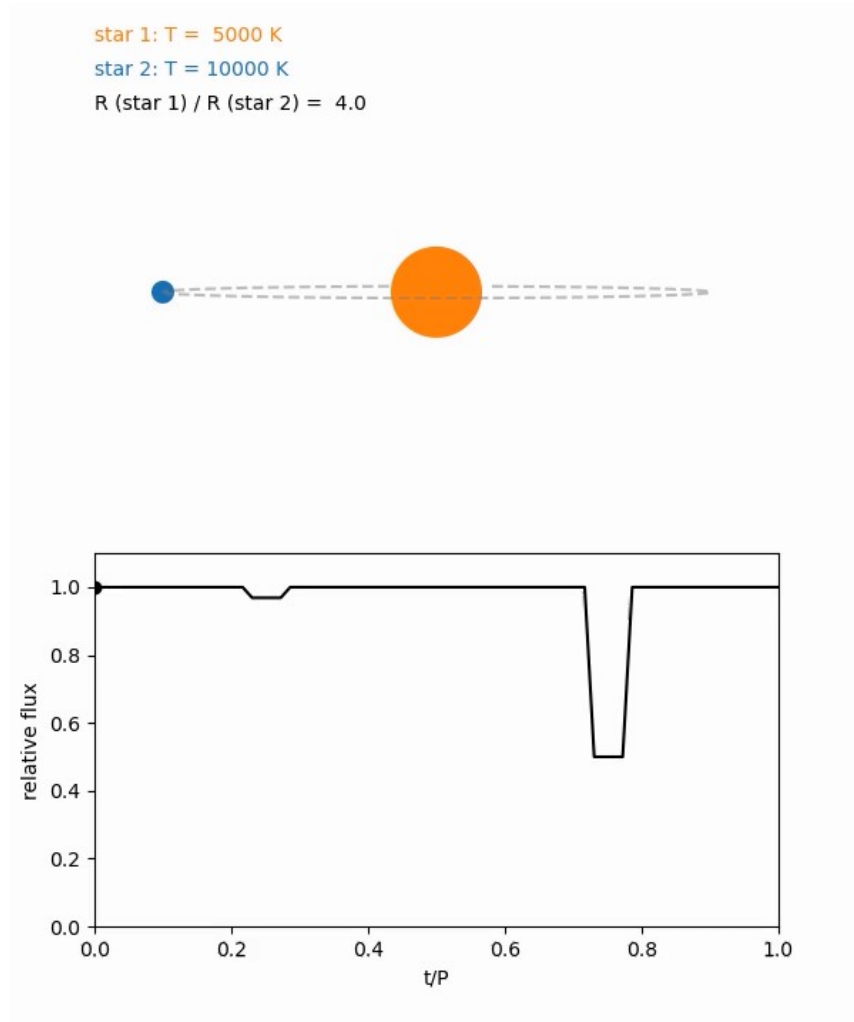


FIG. 2.—Lick Observatory residual velocities (*filled circles*) for υ Andromedae after removal of the Keplerian wobble caused by the inner companion, using best-fit orbital parameters of $P = 4.6171$ days and $K = 75 \text{ m s}^{-1}$. These residual velocities are not random. Two timescales are apparent at 3 and ~ 0.7 yr, the latter especially apparent from 1998.5 to 1999.4 (*bottom right*). Solid line shows the theoretical velocity curve caused by the outer two companions.

Upsilon Andromeda system (Butler et al. 1999)

Transit Method



- Period from time between successive transits
- Radius from dimming fraction + parent star's properties
- Composition: spectra during and outside of transit
- Surface T: IR observations—look for drop in flux
- Limitations
 - Biased toward short periods
 - Mass not measured
 - Needs favorable inclination

Transit Method

Sing et. al: Detection of potassium in XO-2b from GTC spectrophotometry

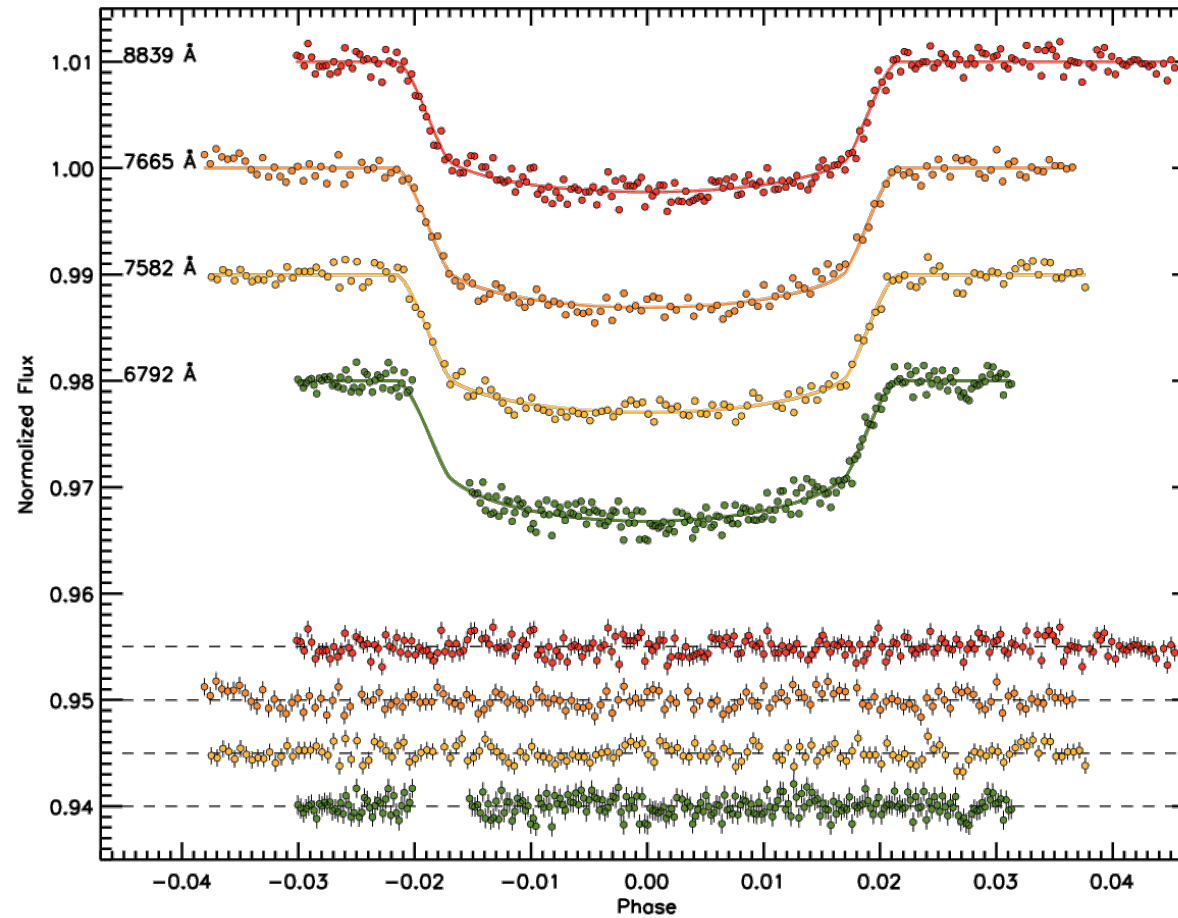
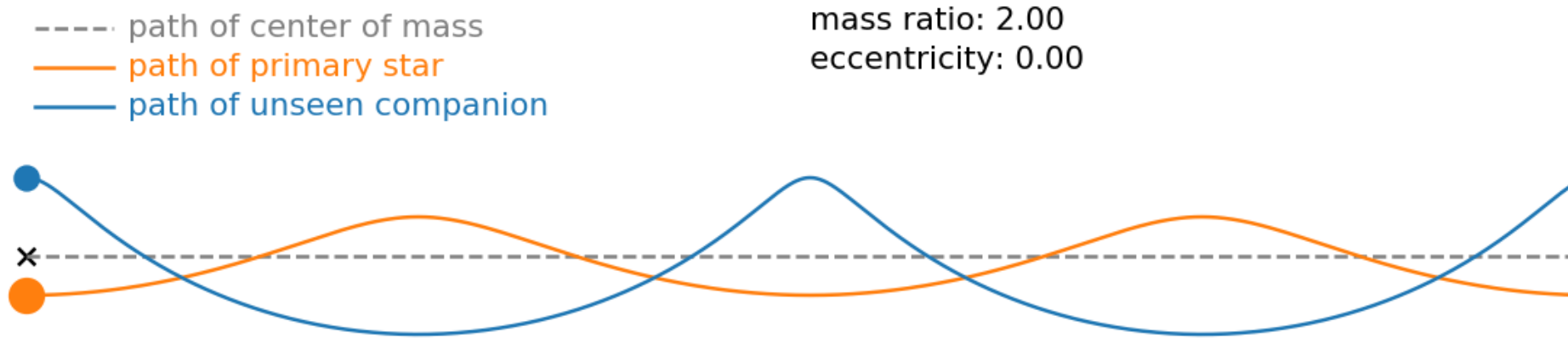


Fig. 2. GTC OSIRIS narrow-band transit light curves at (top-to-bottom) 8839, 7665, 7582, and 6792 Å, with an arbitrary flux offset applied to each curve for clarity. The best-fit models are also over-plotted, along with the residuals (bottom four curves) which have a y-axis error bar corresponding to the per-point flux uncertainty. The light curves are near-Poisson limited, with per-point S/N levels between 1149 and 1452.

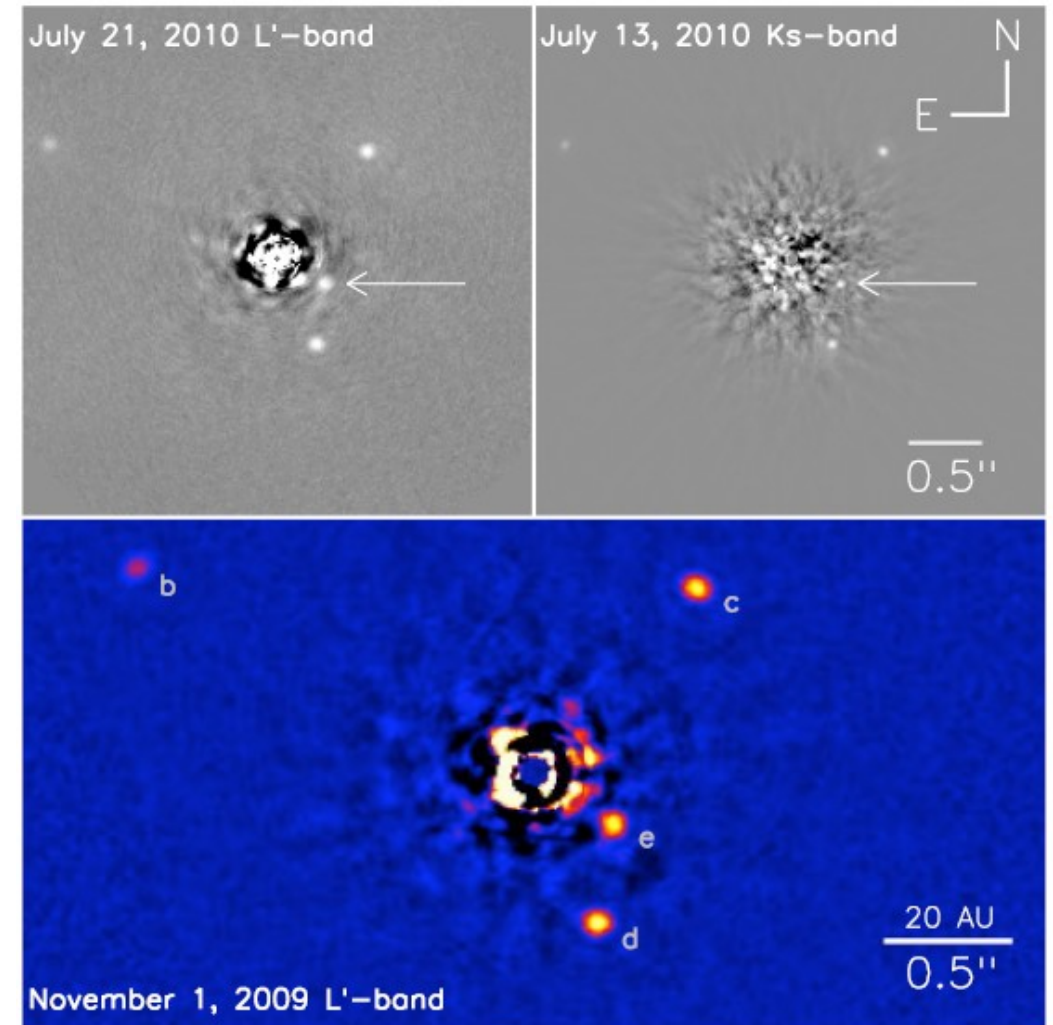
Astrometric Technique



- Lool
- Works best for:
 - Massive planet
 - Larger orbit (but requires longer observations)
 - Nearby stars

Direct Detection

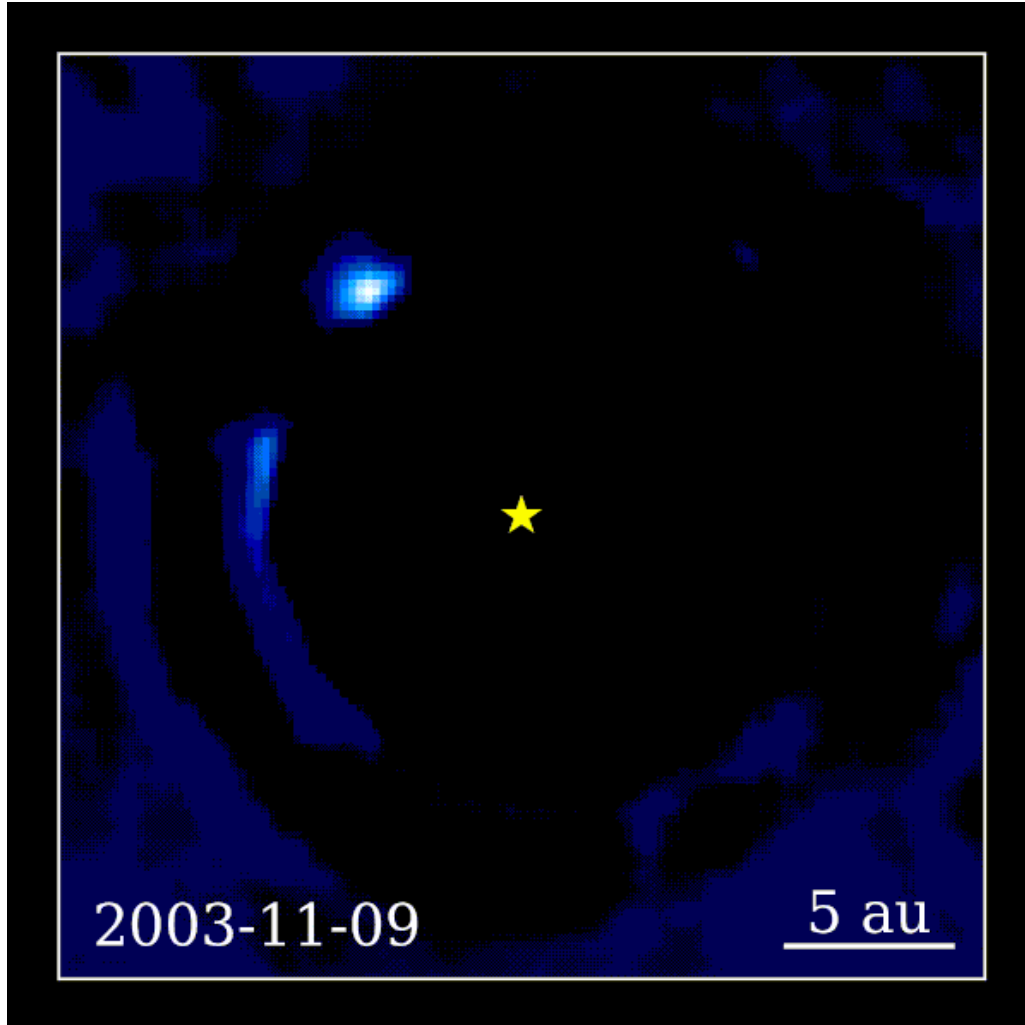
- High contrast imaging
- IR helps---biased toward young planets
- Very few exoplanets discovered this way



HR 8799 (Marois et al. 2010, Nature)

Direct Detection

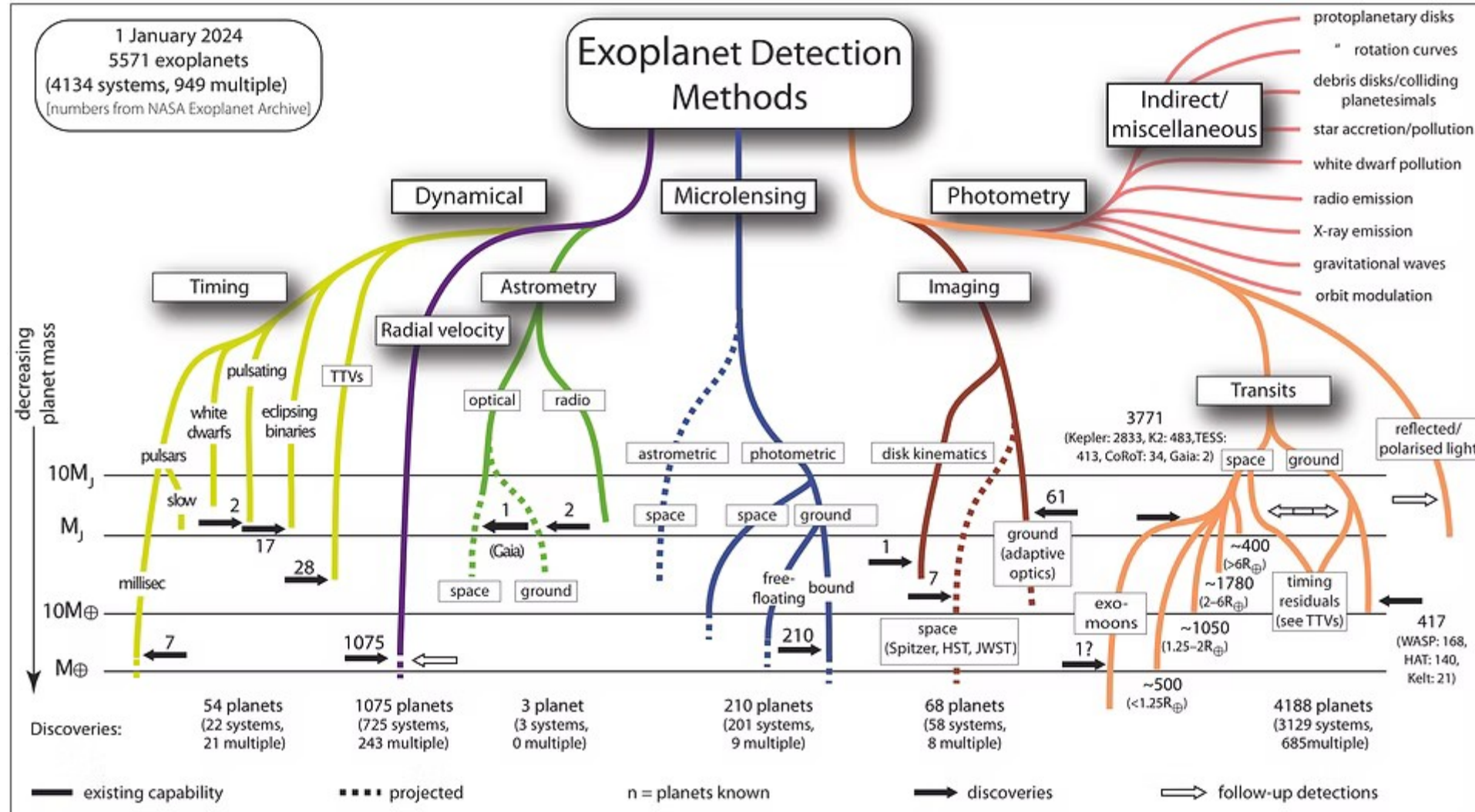
- β Pic system has been imaged over a full period



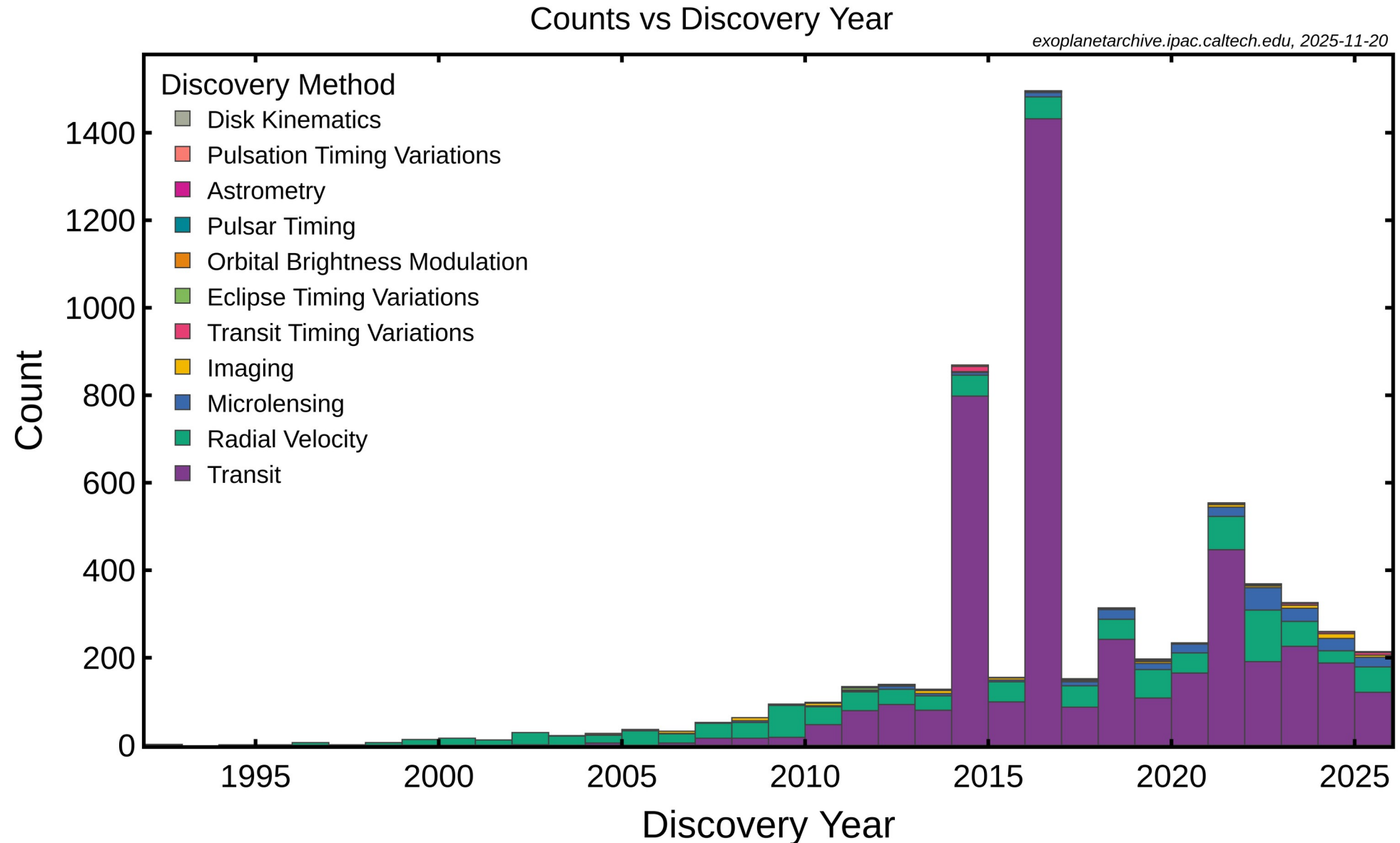
<https://jasonwang.space/orbits.html>

Credit: Jason Wang & Malachi Noel (Northwestern)

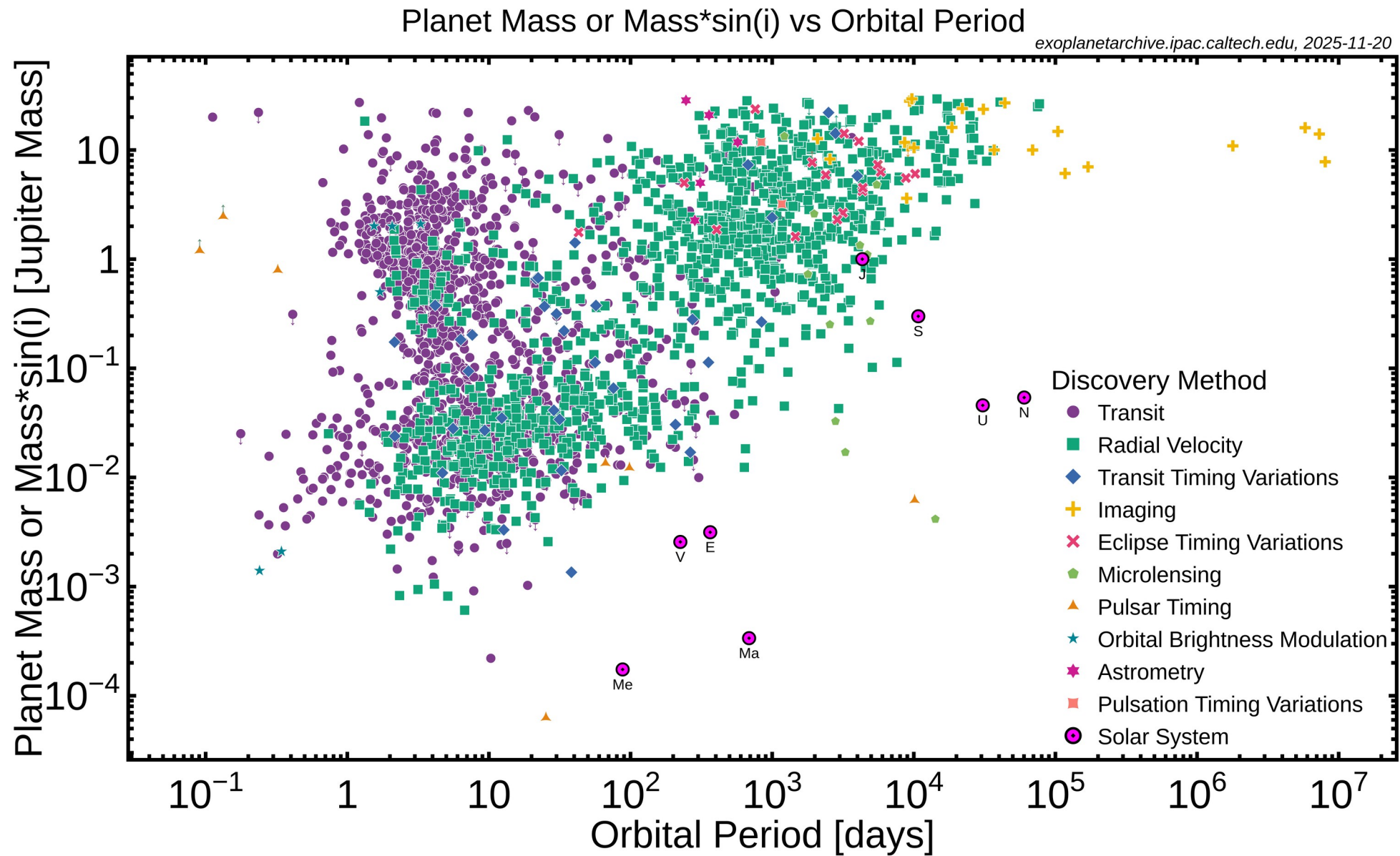
Exoplanet Detection Methods



Exoplanets



Exoplanets



Exoplanets

